

**ASSESSMENT OF THEMATIC MAPPING
TECHNIQUES FOR POPULATION DISTRIBUTION
VISUALIZATION: A Comparative study of Dot Density,
Filled Isopleth and Unfilled Isopleth**

Master Thesis

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Sciences of the Paris-Lodron-University of Salzburg

Submitted by

Babafemi Sunday Olatunji

Supervisor:

Dr. Christoph Traun

Department of Geoinformatics Z_GIS

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Author's Declaration

I hereby declare that this thesis is the result of my own work. All sources used in this work have been properly acknowledged and cited in accordance with scientific and academic ethics.

A handwritten signature in black ink, appearing to be 'Salzburg', written in a cursive style.

(Salzburg, June 2026)

Abstract

Thematic maps play a critical role in communicating spatial patterns and supporting geographic decision making. Among the various methods used to represent population density, Dot Density maps, Filled Isopleths (Colour-graded), and Unfilled Isopleths (Contour-based) are widely applied, yet their influence on users' perception and interpretation of spatial density patterns remains insufficiently understood. This study investigates how these three cartographic representations affect the perception of population density, with particular emphasis on hotspot identification and spatial gradient interpretation. A between-subjects online experiment was conducted using maps representing both real-world and synthetic population distributions. Participants completed tasks while measure of accuracy, response time, and perceived difficulty were collected. Demographic information and self-reported cartographic experience were also recorded.

The findings demonstrate that thematic map design substantially affects the interpretation of population density patterns. Filled Isopleths appear to provide the most effective support for both hotspot identification and gradient interpretation, while Dot Density maps remain useful for rapid exploratory assessment of spatial clustering. These results contribute to cartographic perception research and provide practical recommendations for selecting thematic map types in population mapping and related spatial analysis applications.

Keywords: Cartographic perception, Thematic Maps, Population Density, Filled isopleth, Unfilled Isopleth, Dot Density, Real-World, Synthetic, Hotspot Identification, Spatial Gradients, Map Usability, Spatial Cognition.

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Abbreviations

DD	Dot Density
FI	Filled Isopleth
UI	Unfilled Isopleth
SYN	Synthetic
RW	Real-World
GD	Gradient
HS	Hotspot

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1.Introduction

Thematic maps are special types of maps that show information about a topic, such as population, income, or temperature. They are important in cartography to visualize spatial patterns and support analysis (Slocum et al., 2023).

The visualization of spatial phenomena plays a critical role in understanding geographic patterns and support decision making processes. Thematic maps are widely used to communicate complex spatial information effectively through appropriate symbolization and map design (Slocum et al., 2023). In the context of natural hazard assessment, thematic mapping techniques can be used to represent population distribution built-up areas, and hazard exposure. Ehrlich et al., (2018) demonstrated the importance of integrating population density and built-up area datasets derived from remote sensing to quantify exposure to natural hazards to multiple spatial scales. Several mapping techniques are commonly used to represent population distribution. Dot Density maps use repeated dots to represent quantities within geographic space, enabling users to visually identify clustering, concentration, and dispersion patterns. Filled Isopleth represent continuous spatial variation through graduated colour surfaces or heatmap-style visualization, allowing smooth transitions between density values. In contrast, Unfilled Isopleth, commonly referred to as isolines or contour maps, use lines connecting points of equal value to represent spatial gradients and continuous phenomena without filled colour regions. Unlike discrete spatial representations, isopleth-based visualization estimates values continuously across geographic space (Traun et al., 2017; He et al., 2011; Slocum et al., 2023). All three techniques are valid; each emphasizes different spatial characteristics and may lead to distinct interpretations by map readers. Previous studies have shown that thematic map design can influence user effectiveness, attractiveness ratings, and overall evaluation of spatial information. Furthermore, different visualization methods may affect how spatial patterns are perceived and interpreted, highlighting the importance of selecting an appropriate mapping technique for a given task. Selecting the most appropriate method therefore plays a crucial role in thematic design (Medynska-Gulij et al., 2020; Besançon et al., 2020; Forrest & Medynska-Gulij, 2021).

The effectiveness of these thematic mapping techniques is closely related to cartographic communication and spatial cognition, as different visualization methods requires user to interpret spatial information through varying levels of abstraction and visual complexity (MacEachren 2004). However, different thematic mapping techniques emphasize different

characteristics of spatial data. Because these visualization methods communicate information in different ways, they may influence how users perceive, interpret, and evaluate spatial patterns.

1.1 Map Interpretation and User-Centred Design

Research has shown that the way information is represented on a map influences not only what users see but also how they understand and interpret spatial relationships (MachEachren, 2004). Thematic maps communicate spatial information through abstraction, classification, and symbolization, enabling complex geographic phenomena to be represented in a form that facilitates pattern recognition and interpretation (Ding & Meng, 2014). Modern cartography therefore adopts a user-centred perspective, recognising that perceptual and cognitive factors play a key role in map use and interpretation (Roth et al., 2017).

Recent studies have demonstrated that users do not always interpret maps in the same way. Map-reading performance is influenced not only by the visual characteristics of a map but also by the characteristics of the user and the task being performed. Cognitive and usability studies have shown that different map designs can affect the accuracy and efficiency with which users extract and interpret spatial information (Pickle, 2004). Furthermore, no single map design is optimal for all users or analytical task, highlighting the importance of evaluating cartographic visualizations from a user-centred perspective (Pickle, 2004). Similarly, Medynska-Guly et al. (2023) showed that user testing offers important insights into how well thematic maps work in practice, highlighting the gap between what users like and what helps them perform better.

The findings of Beitlova et al. (2020) further demonstrate that cartographic design can significantly affect map comprehension and task success. Their study showed that users experienced varying levels of difficulty across thematic map types, with performance influenced by factors such as symbol interpretation, legend use, and visual complexity. The authors emphasized that effective map design should support users in identifying patterns, locating information, and completing analytical tasks accurately and efficiently. Consequently, evaluating different thematic mapping techniques through measures such as accuracy, response time, and user perception is essential for understanding their effectiveness in communicating spatial information.

These studies collectively suggest that it is important to carefully manage how much visual information and complexity a thematic map contains. A good map should present enough detail to be useful while remaining clear and easy to understand. This helps people to complete tasks

like finding locations, making comparisons, or spotting patterns without getting confused. By testing and comparing different type of maps, researchers can figure out which designs provide the best mix of accuracy and user satisfaction.

1.1 Research Gap

While several studies have examined thematic mapping principles and cartographic visualization techniques, relatively few have conducted comparative user-based evaluations involving Dot Density, Filled Isopleth (colour graded), and Unfilled Isopleth (contour lines) maps within the same experimental framework. Recent research has demonstrated that the choice of thematic map type can significantly influence map-reading performance and spatial interpretation. For instance, Sukraini et al. (2022) investigated the effectiveness of choropleth and graduated symbol maps for communicating COVID-19 case distributions in Bali, Indonesia. A total of 120 participants completed eleven map-reading tasks, including identifying, comparing, interpreting, categorizing, and locating spatial information. The study evaluated user performance based on three metrics: accuracy response time, and level of difficulty. Results showed that participants using choropleth maps achieved higher accuracy (73%) compared with those using graduated symbol maps (63%). In addition, choropleth maps resulted in faster task completion times, with an average response time of 28.15 seconds and compared to 35.78 seconds for graduated symbol maps. Participants also perceived choropleth maps as easier to read and understand. Based on these findings, the authors concluded that choropleths maps provided more effective means of communicating spatial patterns and supporting map-reading tasks than graduated symbol maps. Similar findings were reported by Słomska-Przech and Gołębiowska (2021), who compared choropleth, graduated symbol, and isoline maps using a large-scale user study involving 366 participants. Participants completed eleven map-reading tasks designed to assess several aspects of spatial information processing, including identification, comparison, interpretation, and pattern recognition. The study measured answer accuracy, completion time, and perceived task difficulty. The results indicated that choropleths maps achieved the highest overall accuracy required the shortest response times and were perceived as the easiest map type to use. In contrast, isoline maps generally produced lower accuracy, longer responses times, and higher difficulty ratings. The authors concluded that thematic map type plays a significant role in influencing both objective performance and subjective user experience, and that certain map representations are better suited to particular analytical tasks than others.

Together, these studies provide strong evidence that cartographic representation influences how users interpret and understand spatial information. However, previous research has primarily focused on comparisons involving choropleth, graduated symbol, and isoline. As a result, the relative effectiveness of Dot Density, Filled Isopleths, and Unfilled Isopleths maps for representing spatial distribution patterns under both controlled synthetic and realistic conditions remains unclear. Therefore, this study extends existing cartographic perception research by comparing these thematic map types using objective performance measures (accuracy and response time) as well as subjective evaluations of perceived difficulty.

1.2 Aim of the study

This research aims to fill that gap by testing these three map types using both real-world population data and simulated datasets. It examines how each technique affect user' ability to recognize population patterns such as hotspot and gradient effectively.

1.3 Research Objectives

1.4.1 Main Objective

To assess the impact of different thematic mapping techniques on the interpretation of spatial population distribution patterns.

1.4.2 Specific Objectives

1. To compare the effectiveness of Dot Density maps, Filled Isopleths, and Unfilled Isopleths in hotspot identification tasks.
2. To evaluate how different thematic mapping techniques support spatial gradient interpretation along continuous population profiles.
3. To compare interpretation accuracy and response time across thematic map types.
4. To assess perceived cognitive difficulty associated with hotspot and gradient tasks.
5. To investigate the relationship between subjective task difficulty and objective interpretation performance.
6. To evaluate thematic map usability using both synthetic and real-world population datasets.
7. To provide recommendations for selecting appropriate thematic mapping techniques for population visualization and spatial decision-making tasks.

1.5 Document organization

In the following sections, I describe the design of the study, including the data sources, the simulation process, and the methods used to create and evaluate thematic maps. Then, I present the results of the comparative analysis of the map techniques, focusing on user performance. Finally, I concluded with the main outcomes of the study and offer recommendations.

2. Methods

This study employs a controlled web-based user experiment to evaluate the effectiveness of three thematic mapping techniques; Dot Density (DD), Filled Isopleth (FI), and Unfilled Isopleth (UI) in supporting the interpretation of spatial population distribution patterns. The research focuses on two primary spatial interpretation tasks: hotspot identification and spatial gradient interpretation. To assess map effectiveness, the experiment measures interpretation accuracy, response time, and perceived task difficulty. These metrics correspond to the commonly applied usability performance metrics of effectiveness (correctness), efficiency (time), and subjective task difficulty, which are widely used in cartographic usability and thematic map evaluation studies (Çöltekin et al., 2009; Słomska-Przech and Gołębiowska, 2021)

2.1 Experimental Design

A between-subjects experimental design was adopted to investigate performance difference among the three thematic mapping techniques. In alignment with user-centred design frameworks in cartography usability (Roth, 2013), participants were exposed to only one cartographic representation throughout the survey. The approach minimizes potential learning effects, prevents for carry-over effects, visual familiarity effects, and participant fatigue that may occur when users interact with multiple visualization conditions during same experiment (Lazar et al., 2017). This structure ensures that performance variances can be directly attributed to the cartographic design itself rather than past task exposure (Roth, 2013).

The between-subjects experimental framework was considered appropriate because the study focuses on comparing cognitive interpretation differences between distinct cartographic representations under controlled conditions.

The independent variable (IV) in this study is the thematic visualization techniques with three experimental conditions:

- Dot Density maps (DD)
- Filled Isopleth/Colour graded maps (FI)
- Unfilled Isopleth/Contour lines maps (UI)

Participants were randomly assigned to one of the three experimental groups using a randomization procedure implemented within LimeSurvey.

The dependent variables (DVs) measured in the study include:

- Interpretation Accuracy: This binary metric (0=incorrect, 1=correct) captures a participant's ability to correctly identify hotspot regions and accurately interpret gradient patterns within thematic maps.
- Response Time: Recorded as the exact duration (In seconds) between the initial map stimulus presentation and participant submission of an answer. This temporal metric serves as an objective measure of cognitive efficiency and processing speed during map-reading workflow.
- Perceived Difficulty: Evaluated using a self-reported 5-point ordinal Likert scale administered immediately after each task. This subjective metric measures user-perceived cognitive workload, capturing how easy or difficult participants found the hotspot and gradient interpretation processes.

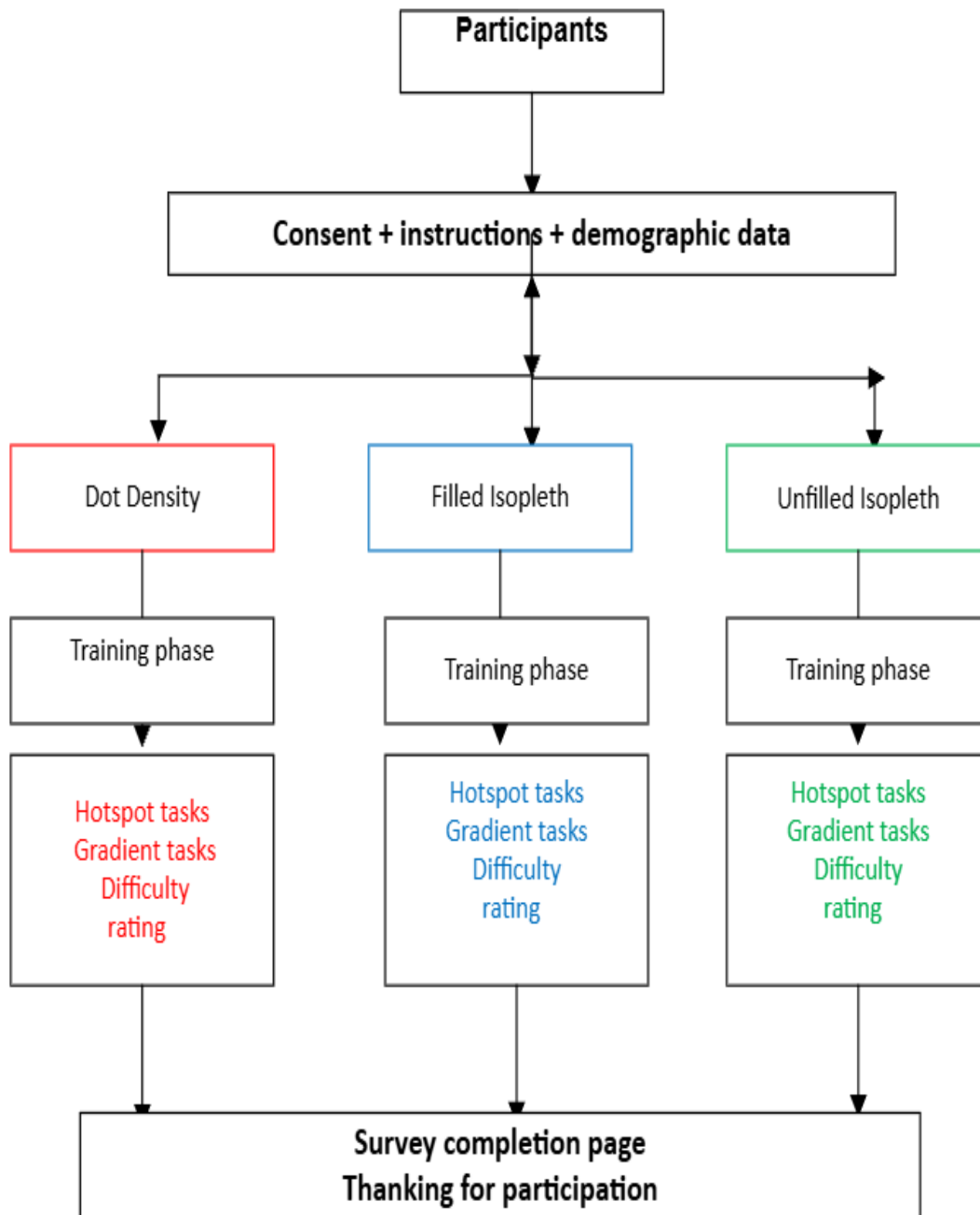


Figure 1: Graphical overview of the experimental workflow and random assignment procedure used on Limesurvey.

2.2 Stimuli Generation

To balance real-world relevance with tight experimental control, I used both actual real-world population data and synthetic population datasets to create the maps. Every thematic map variant was built using these exact same underlying datasets. I kept the layouts, boundaries, scales, and visual frameworks completely identical across all three mapping styles. The strict standardization isolates the mapping techniques themselves, eliminating outside design biases so we can purely measure the different map styles perform against each other.

2.2.1 Real-World population Datasets

Two real-world study areas were selected to represent contrasting spatial population patterns. The Ruhr Region in Germany was used to represent a density gradient due to its highly urbanized and polycentric structure, where multiple cities merge into a continuous urban landscape with smooth transitions toward surrounding rural areas. Selected regions included Essen, Dortmund, Bochum, Duisburg, Gelsenkirchen, Recklinghausen, and several neighbouring districts. Population data were obtained from the worldPop database, while administrative boundaries were derived from GADM database.

In contrast, California, United States, was used to represent highly clustered population hotspot patterns characterized by discrete high-density urban concentrations separated by lower-density surrounding regions. This dataset was primarily used for hotspot identification tasks and was also sourced from the same database.

Real-world population datasets were used to evaluate thematic map interpretation under realistic geographic conditions.

2.2.2 Synthetic Population Datasets (Controlled Spatial Patterns)

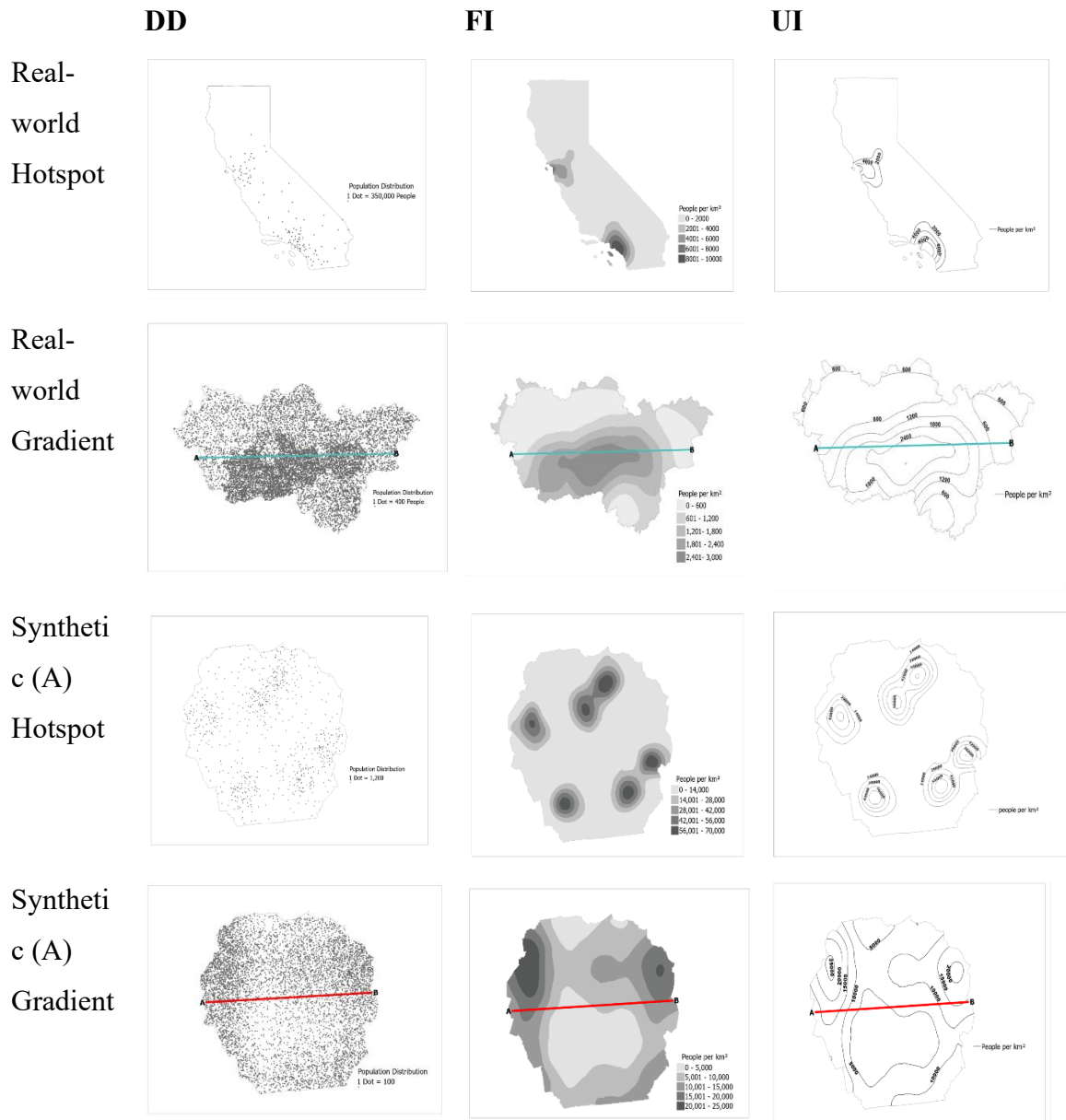
Synthetic population datasets were generated using procedural simulation techniques implemented in Python. AI-assisted coding tools were used to support portions of the scripting and workflow. These datasets were generated to create controlled distribution patterns that isolate specific cartographic interpretation challenges. The synthetic approach enabled precise control over cluster density, spatial arrangement, gradient transition, and distribution complexity. Four primary synthetic spatial structures were generated: two for clustered distribution and two for density gradients. The datasets were created within the artificial polygonal basemap adapted from previous cartographic visualization research (Traun., 2021), providing a neutral spatial framework for the generation of synthetic population patterns.

Population distributions were represented as individual points. This controlled approach ensured that identical underlying spatial phenomena could be transformed consistently across the three thematic mapping techniques. The use of synthetic datasets improves experimental control and allows direct comparison of thematic map performance under predefined spatial conditions.

2.2.3 Cartographic Implementation

To ensure fairness across visualization conditions, all thematic maps were derived from identical underlying datasets and processed using consistent cartographic parameters, allowing interpretation differences to be attributed primarily to the thematic representation itself rather than underlying data variation. Dot density maps were generated using repeated point symbols distributed spatially according to population concentration patterns. Dot values and symbol sizes were standardized to maintain visual comparability with the continuous surfaces. Point-based population datasets were transformed into continuous surfaces for the generation of filled and Unfilled Isopleths using Kernel Density Estimation (KDE). Filled isopleth maps were rendered using continuous grayscale colour gradients to represent variations in population density across geographic space. A neutral Gray-light-to-dark colour scheme was intentionally adopted to minimize visual dominance and reduce potential perceptual bias relative to the other thematic mapping techniques used in the study. Unfilled Isopleths were generated using contour lines representing equal density intervals across the interpolated surfaces. To maintain visual consistency and comparability with the Filled Isopleth maps, the upper-class values from the Filled Isopleth classification scheme were used as contour intervals for the Unfilled Isopleths. This approach reduced excessive contour line density, improved map readability, and ensured fair visual comparison across the different thematic mapping techniques used in the study. To maintain consistency and fairness throughout the experiment, all thematic maps were standardized using: identical map extents, consistent layout design, similar visual hierarchy, consistent legend positioning, and uniform display conditions.

Total of 18 maps were created and served as the stimuli to be used when solving different map use task (Figure 2).



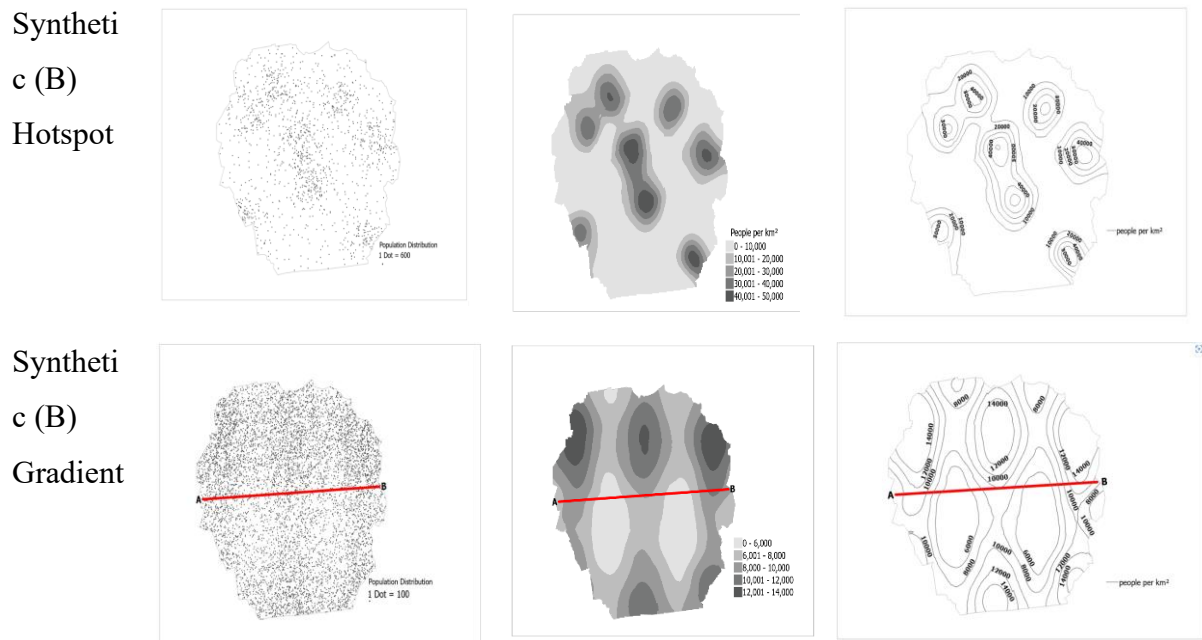


Figure 2: Map types assessed: dot density map (DD), filled isopleth map (FI), Unfilled isopleth map (UI)

2.3 Task Taxonomy

All participants performed identical spatial interpretation tasks regardless of their assigned visualization group. The experimental tasks were designed to evaluate two core aspects of thematic map interpretation: hotspot recognition and spatial gradient interpretation.

2.3.1 Hotspot Identification Task

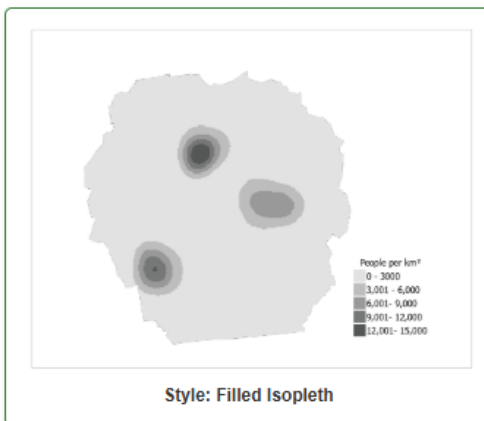
*“In this study, the term **hotspot** is used to describe areas of high population distribution that visually stand out from their surroundings. Although the term **clusters** may be more appropriate from a spatial analysis perspective, **hotspot** was retained because it was used consistently throughout the survey design, task descriptions, stimulus labels, and subsequent analyses. The term is therefore used in an operational sense to refer to visually identifiable concentrations of population rather than statistically derived hotspots.”*

The hotspot identification task evaluated participants' ability to recognize areas of high population concentration within thematic maps. Participants were instructed to determine the number of distinct hotspot regions visible on each map stimulus. This task measures cluster recognition, density concentration perception, and visual pattern interpretation across the three

thematic mapping techniques. Instructions and specific practice examples were assigned to each of the thematic map type.

Example Task: Identifying Hotspots (Filled Isopleth)

In this section, you will use **Filled Isopleth Maps**. These maps use shaded areas to represent different levels of density.



How to read this map:

- **Hotspots:** Look for the **darkest shaded areas**. These represent the highest concentrations.
- **Low Density:** Lighter or white areas represent lower values.
- **Goal:** In the example above, there are **3** distinct dark regions (hotspots).

Observe how the color gradients lead to the center of the hotspots.
When you are ready, click **Next** to proceed.

Choose one of the following answers

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5

Figure 3: One of the Practice examples for hotspot identification

2.3.2 Spatial Gradient Interpretation Task

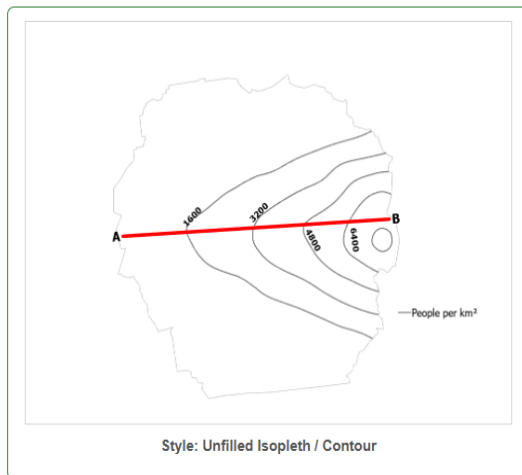
The spatial gradient interpretation task evaluated participants ability to interpret continuous changes in population density across geographic space. Participants were asked to observe density changes along a predefined transect line extending from point A to B and then select a graph that best represented the observed density variation. The task evaluates spatial reasoning, gradient interpretation, and continuous surface perception.

Example 2: Matching the Profile Chart (Unfilled Isoleth)

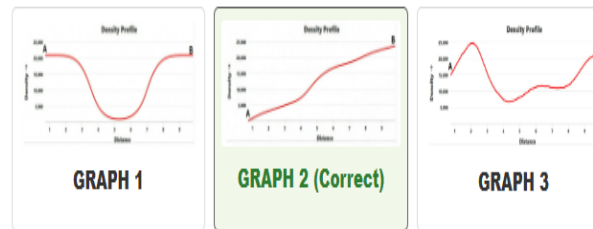
Task: You will see a map with a horizontal line (transect) from **A** to **B**. You must choose which chart correctly illustrates the change in population density along that line.

- **Reading the Contour Map:** Moving from left to right (**A** to **B**), observe the lines. When lines are **closer together** or form **nested loops**, the density is increasing.
- **Reading the Charts:** The vertical axis (**Y**) represents density; the horizontal axis (**X**) represents the distance from **A** to **B**.

Observe the Contour/Line Gradient from A to B:



Which graph correctly matches the increase from A to B shown above?



Why is Graph 2 correct for Unfilled Isoleth?

The map shows few lines at **A**, but as you move toward **B**, the lines become **more frequent** and eventually form a small, tight circle at the peak.

✗ **Graph 1:** Incorrect. This would imply the line frequency drops in the middle, but our map shows lines getting closer together consistently toward **B**.

✓ **Graph 2:** Correct. It represents the "uphill" climb in density as the contour lines get tighter.

✗ **Graph 3:** Incorrect. This would show multiple separate "peaks" of nested circles, but we only have one main peak at **B**.

Figure 4: One of the practice examples for gradient interpretation

2.3.3 Subjective Difficulty Assessment

Following each experimental task, participants were asked to evaluate how easy or difficult they found the task to perform. The subjective difficult assessment was included to measure perceived cognitive workload and thematic map usability across visualization techniques.

These responses were later compared with objective performance measure such as interpretation accuracy and response time.

*

**Thinking about the map you just viewed:
How easy or difficult was it to perform the task?**

Choose one of the following answers

- ☐ Very easy
- ☐ Easy
- ☐ Neutral
- ☐ Difficult
- ☐ Very difficult

Figure 5: Difficulty level task

2.4 Participants and Data

The main study was conducted through Limesurvey distributed via academic networks and social media platforms. Invitations were shared with university students, geoinformatics-related communities, and personal contacts with the request for further dissemination. To improve participant diversity and achieve more balanced experimental groups, additional responses were collected through Clickworker platform. Online crowdsourcing platforms have been widely used in cognitive behavioural, and usability research due to their ability to provide diverse participants samples and reliable experimental data (Crump et al., 2013; Peer et al., 2017). Most specifically, Clickworker has previously been employed in cartographic and geovisualization research, demonstrating its suitability for web-based user experiments (Traun et al., 2021). A total 401 responses were collected before data cleaning. To improve data quality and reduce the influence of careless or unrealistic submissions, several quality-control procedures were applied. Incomplete responses, unrealistically short completion times, and suspicious response patterns were removed from the dataset. In addition, unusually long duration responses were manually inspected to distinguish inactivity from careful participation. Response consistency, task completion behaviour, and overall answer quality were also considered during the cleaning process. Following the cleaning procedures, the final dataset consisted of 178 valid responses distributed across the three experimental groups. The DD group contained 58, the FI group contained 62, and the UI group contained 58 participants, respectively.

The final sample showed balanced gender distributions across the experimental groups. The DD group consisted of 29 male and 29 female, the FI group included 32 male and 30 female, while the UI group consisted of 37 male and 21 female participants. See figure 6 for distribution of age and gender by group and see Table 1 and 2 for self-reported cartographic experience and educational level.

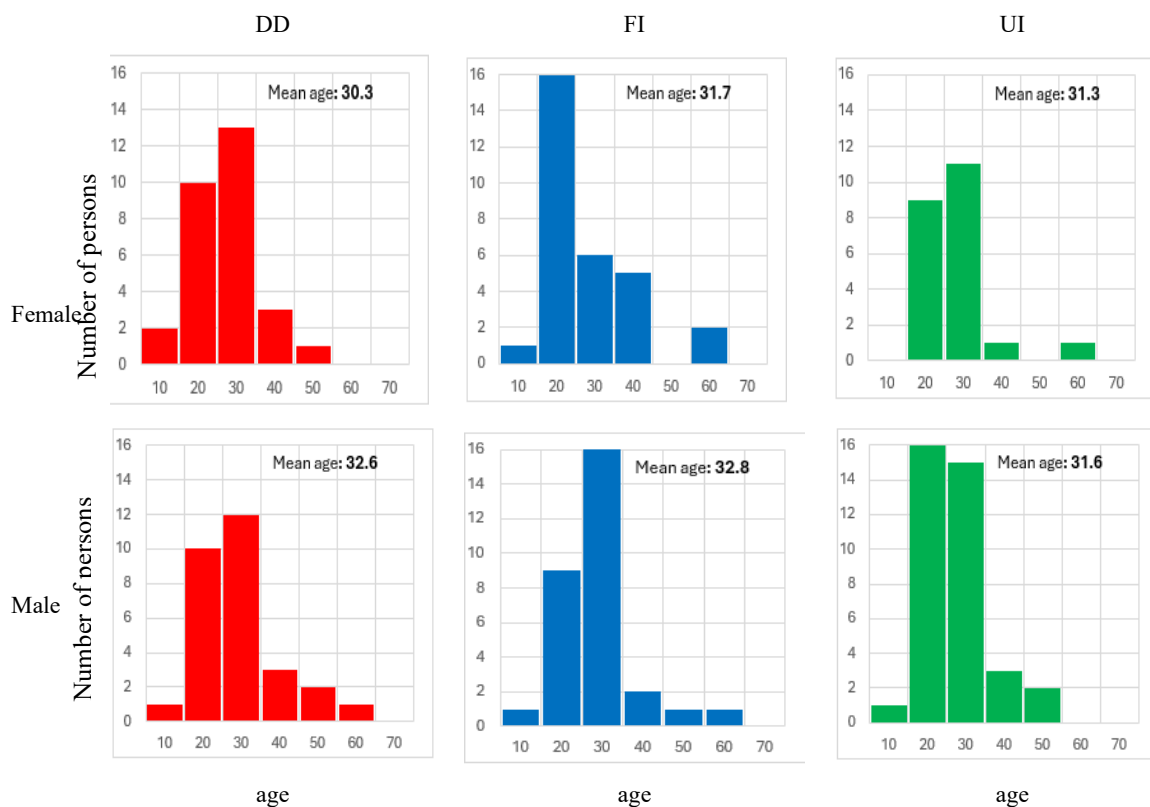


Figure 6: Age distribution and group

Table 1. Participants educational level (Raw count)

	DD	FI	UI	Σ
High School	9	10	4	23
Bachelor's Degree	27	31	30	88
Master's Degree	17	15	15	47
PhD / Doctorate	3	4	4	11
Others	2	2	5	9
Σ	58	62	58	178

Table 2: Self-reported cartographic experience (Raw count)

	DD	FI	UI	Σ
Beginner	15	7	9	31
Basic	5	5	5	15
Intermediate	15	23	11	49
Advanced	13	15	24	52
Expert	10	12	9	31
Σ	58	62	58	178

2.5 Data Analysis

Preprocessing and thematic map generation were performed using GIS and spatial analysis software including ArcGIS Pro/Python. The collected data were cleaned on Excel using quantitative statistical methods to evaluate differences in thematic map interpretation performance across the three visualization techniques. Descriptive statistics were used to summarize participants' demographics, interpretation accuracy, response times, and perceived difficulty ratings. Comparative statistical analyses were conducted to evaluate performance differences between Dot Density maps, Filled Isopleths, and Unfilled Isopleths. The analysis also examined relationships between subjective difficulty ratings and objective performance measures. Statistical analyses were conducted using Jamovi a free and open-source statistical software based on R programming language. Jamovi was selected because it combines the computational power and reproducibility of R with a user-friendly graphical interface. The software supports a wide range of statistical procedures, making it suitable for quantitative research and academic applications (Navarro & Foxcroft, 2019; Sahin & Aybek, 2020; Silva de Souza et., 2024; Ashour, 2024)

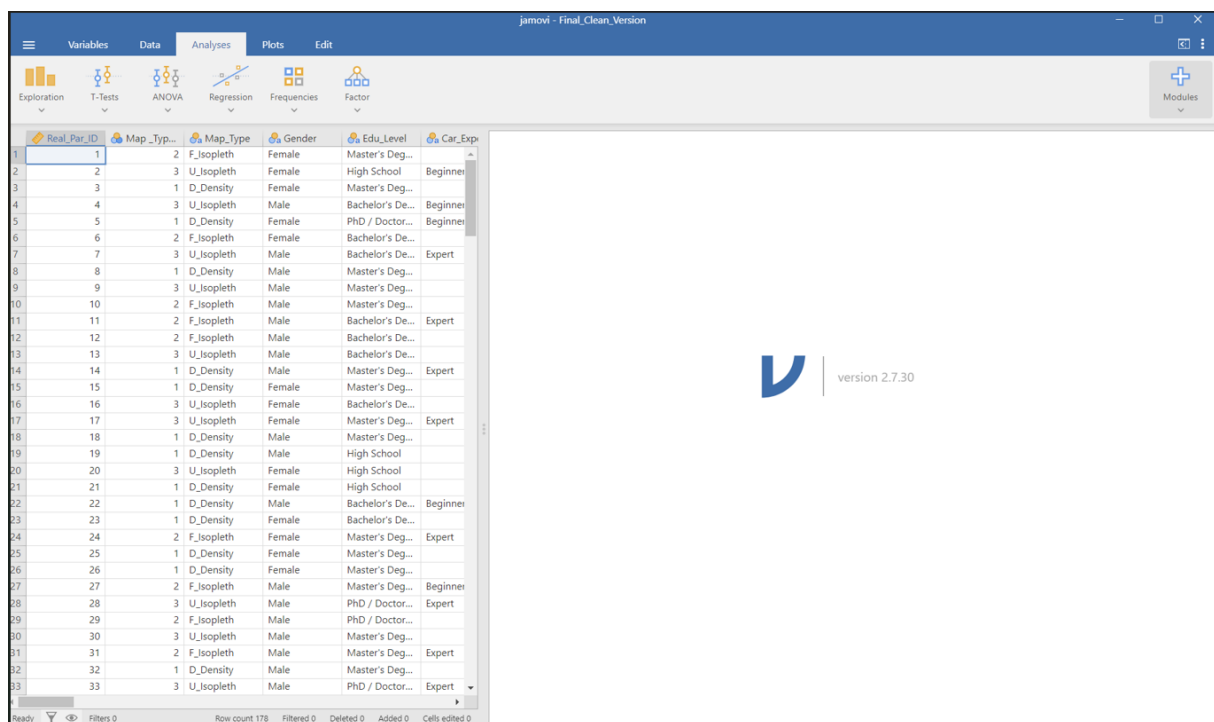


Figure 7: Jamovi (Version 2.7.30) interface displaying the main data view panel.

3. Result

3.1 Accuracy

3.1.1 Descriptive findings

Figure 8 presents the average accuracy scores obtained across the three thematic mapping techniques for both real-world and synthetic spatial interpretation tasks. The results reveal noticeable differences in performance depending on the spatial task and cartographic representation used. For real-world hotspot task (RW_HS) the DD representation achieved the highest average accuracy among the three map types. This suggests that discrete point-based representation may support hotspot localization effectively in real-world spatial distributions by allowing participants to visually identify concentrated clusters more directly. In comparison, FI produced lower accuracy values for this task.

In contrast, the gradient interpretation task using the real-world dataset (RW_GR) showed improved performance for FI. FI achieved the highest average accuracy, while DD and UI produced comparatively lower results. Similar trends were observed in the synthetic hotspot tasks (SYN_HS_A) and SYN_HS_B), where FI consistently outperformed the other mapping techniques. For the synthetic gradient tasks, SYN_GD_A, FI, and DD achieved almost identical and highest accuracies while UI was noticeably lower. However, the differences between the map types were less pronounced in SYN_GD_B, where the overall performance became more balanced across the three representations.

Overall, the results indicate that the effectiveness of thematic mapping techniques varies according to the spatial information task being performed. While DD maps support hotspot identification effectively in certain real-world contexts, FI generally demonstrated stronger performance for gradient interpretation and several synthetic pattern tasks. These findings reinforce the idea that no single thematic representation is universally optimal for all forms of spatial interpretation.

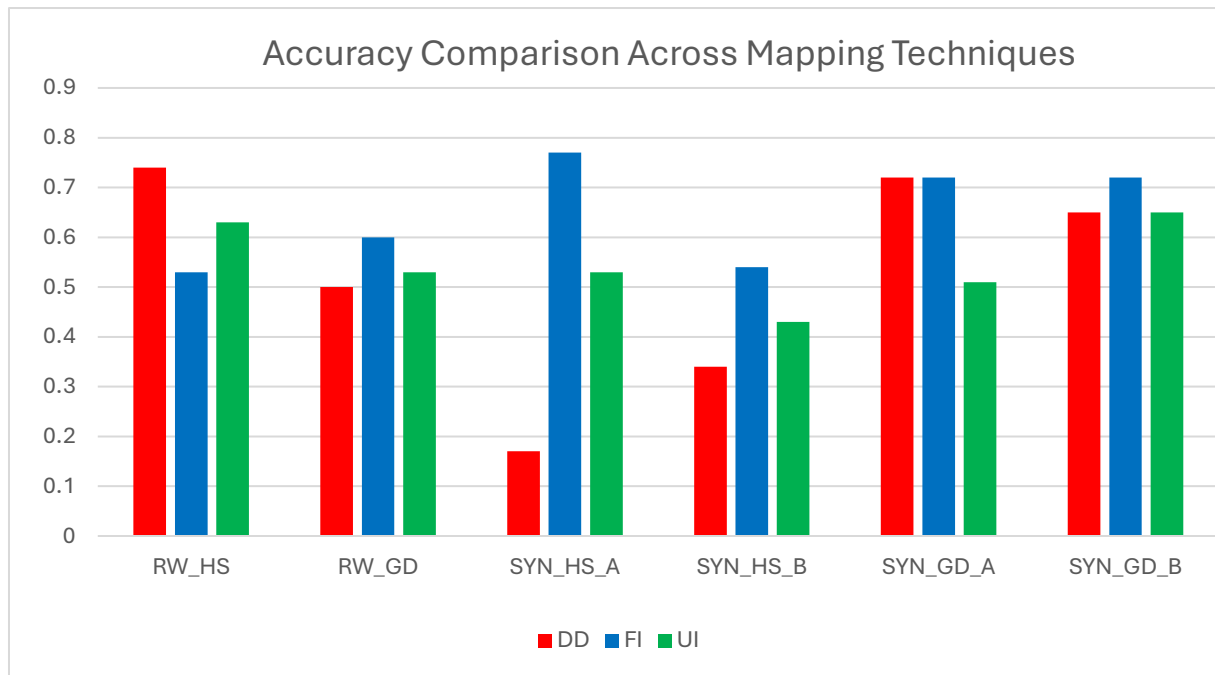


Figure 8: Average accuracy scores across thematic mapping techniques for real-world and synthetic spatial interpretation tasks.

3.1.2 Statistical Significance

Prior to inferential analysis, the distribution properties of the dataset were assessed using histograms, q-q plots, skewness statistics, and the Shapiro-Wilk normality test. The results indicated substantial deviations from normality across several variables.

Accuracy data consisted of binary categories outcomes (correct (1) versus incorrect (0) responses). Therefore, Chi-square tests of independence were applied to examine whether answer accuracy differed significantly across the thematic mapping techniques. Cramer's V was additionally calculated to assess the strength of association between map types and answer accuracy.

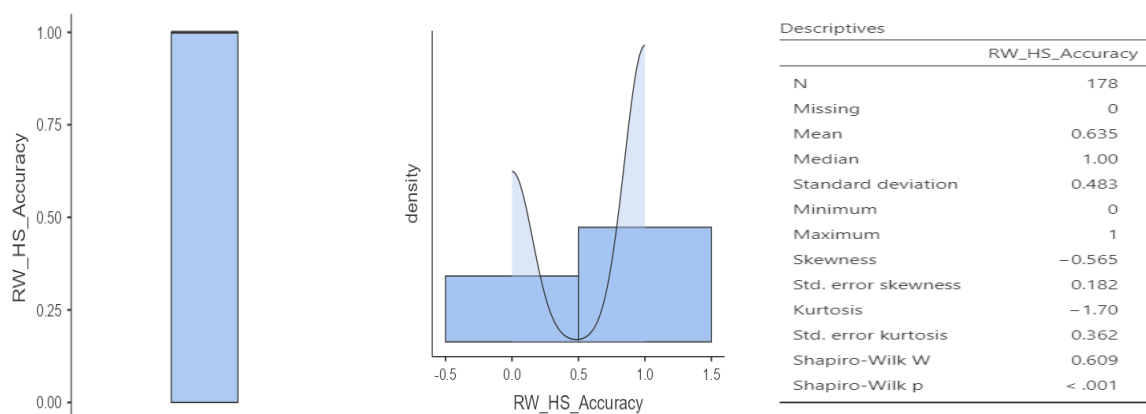


Figure 9: Normality test Accuracy outcome of one of the Variables

The average accuracy for all the 6 tasks was 58%. Filled Isopleths achieved the highest overall accuracy 65%, outperforming both Dot Density maps 52% and Unfilled Isopleths 55%. The results suggest that participants were more successful when interpreting continuous filled density surfaces compare to discrete or contour-based representation.

The accuracy of the answers differed significantly between users of the three different map types: $X^2(2, N = 1068) = 13.3, p = 0.001$, Cramer's $V = 0.112$. Pairwise comparisons showed that the differences in answer accuracy between each pair of the tested map types were statistically significant significantly:

- DD-FI $X^2(1, N = 720) = 12.1, p < .001$, Cramer's $V = 0.130$ (Filled Isopleth approximately 13 percentage points higher than Dot Density)
- DD-UI $X^2(1, N = 696) = 0.578, p = 0.447$, Cramer's $V = 0.29$. (No statistically significant difference in answer accuracy was observed between Dot Density and Unfilled Isopleth)
- FI-UI $X^2(1, N = 720) = 7.33, p = 0.007$, Cramer's $V = 0.101$ (Filled Isopleth maps producing approximately 11 percentage points higher accuracy than Unfilled Isopleth).

To account for repeated observations from the same participant and to further examine the influence of map type and spatial task type on interpretation accuracy, a logistic generalized linear mixed model (GLMM) was fitted with participant ID included as a random intercept. The model revealed a significant effect of map type, $X^2(2) = 7.45, p = 0.024$, indicating that accuracy differed across the thematic map representations. A significant effect of task type was also observed, $X^2(1) = 13.31, p < 0.001$, suggesting that participant performance varied between hotspot identification and gradient interpretation tasks. Furthermore, a significant interaction between map type and task type was found, $X^2(2) = 6.72, p = 0.035$, indicating that the effectiveness of the map types depended on the type of task being performed.

The GLMM indicated that FI representation produced significantly higher accuracy than DD representation ($p = 0.009$), whereas no significant difference was observed between the UI and DD representations ($p = 0.564$). these findings are consistent with the pairwise Chi-square analyses, which showed that the primary difference occurred between FI and DD.

Accuracy of answers differed also between tasks. See figure 10 for average percentage for each task.

The inferential statistical analysis as shown in Table 3 that answer accuracy differed across the tested thematic mapping techniques for several tasks. Chi-square tests of independence were conducted to examine whether the type of thematic map significantly influenced participants' interpretation accuracy. Cramer's V was additionally used to assess the strength of association between map type and task performance. Where statistically significant differences were identified, post-hoc pairwise comparisons were conducted to determine which mapping techniques differed significantly from one another. Statistical significance was evaluated at the 0.05 significance level.

For the Real-world Hotspot task (RW_HS), a statistically significant association was observed between map type and answer accuracy, $p = 0.059$, Cramer's $V = 0.167$. Pairwise comparisons showed that DD:FI ($p = 0.018$) However, no statistically significant difference was found between DD:UI and FI:UI.

Similarly, the Real-World Gradient task (RW_GD) showed no statistically significant relationship between map type and answer accuracy, $p = 0.557$.

For the Synthetic Hotspot A task (SYN_HS_A), a high statistically significant difference in answer accuracy was found among the three map types, $p = < 0.001$, Likewise pairwise comparisons reached statistical significance, Filled Isopleth outperformed both Dot Density and Unfilled Isopleth. DD:FI ($p = < 0.001$), FI:UI ($p = 0.006$). Unfilled Isopleth also received higher statistically significant over Dot Density ($p = < 0.001$)

In contrast, the Synthetic Hotspot B task (SYN_HS_B) shows no statistically significant association between map type and answer accuracy, $p = 0.079$, Pairwise comparisons indicated that Filled Isopleth maps achieved significantly higher accuracy than Dot Density maps (DD:FI $p = 0.025$).

For the Synthetic Gradient A task (SYN_GD_A), Shows statistically significant association between map type and answer accuracy, $p = 0.024$, Pairwise comparisons demonstrated that the success rates were nearly identical for DD:FI $p = 0.984$, therefore, there was not statistically significant. Both Dot Density and Filled Isopleth maps outperformed Unfilled Isopleth maps DD:UI ($p = 0.022$) and FI:UI ($p = 0.018$).

Similarly, the Synthetic Gradient B task (SYN_GD_B) showed no statistically significant association between map type and answer accuracy, $p = 0.629$.

Overall, the findings consistently indicate that Filled Isopleth maps supported higher interpretation accuracy across several hotspot and gradient interpretation tasks. Although the observed effect sizes ranged from small to moderate, the repeated statistical significance across multiple tasks suggests that Filled Isopleths provide a more effective representation for interpreting continuous population density patterns compared to Dot Density and Unfilled Isopleth techniques.

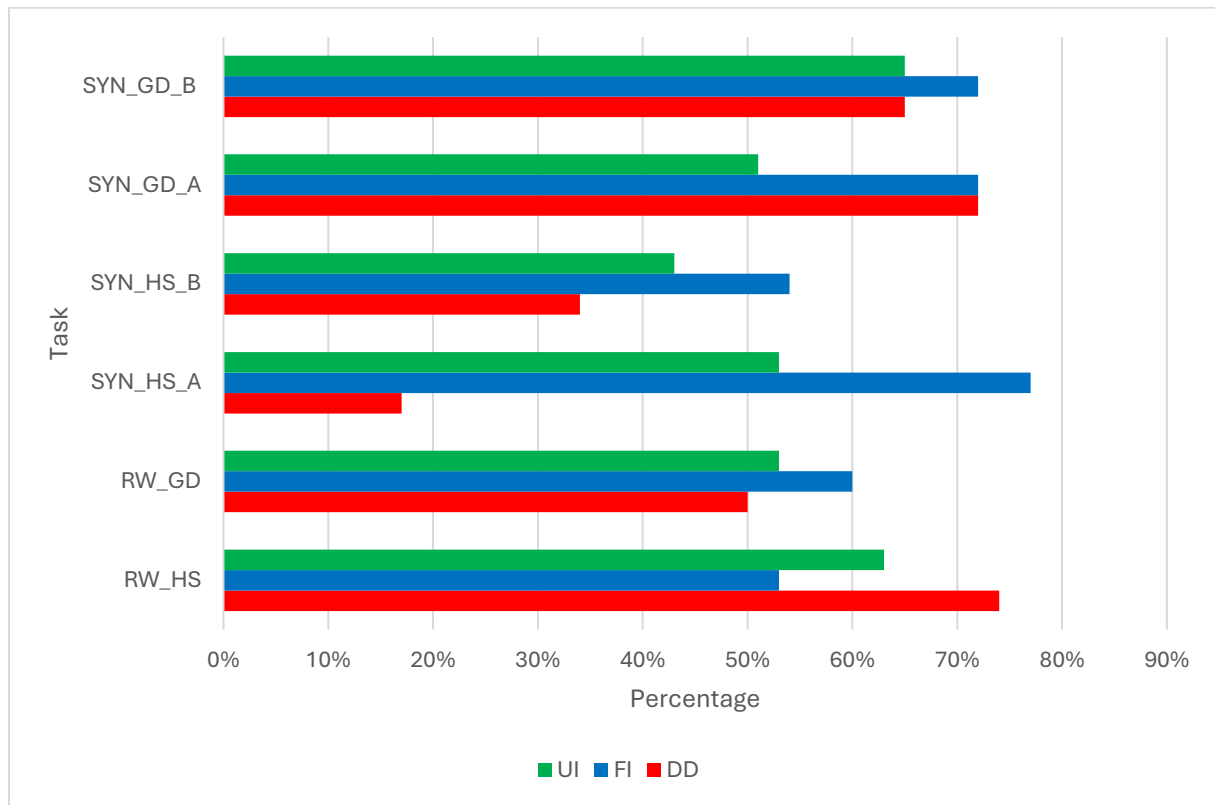


Figure 10: Percentage accuracy scores across thematic mapping techniques for real-world and synthetic spatial interpretation tasks.

Table 3. Inferential Statistical analysis of answer accuracy across the three thematic map types.

Task	Chi-Square Overall χ^2	<i>P</i>	Cramér's V	Pairwise Comparison	χ^2	Cramér's V	<i>P</i>	Significant
RW_HS	5.66	0.059	0.178	DD-FI	5.64	0.217	0.018	Yes
				DD-UI	1.45	0.112	0.229	-----
				FI-UI	1.38	0.107	0.241	-----
RW_GD	1.17	0.557	0.811	DD-FI	0.13	0.0972	0.136	-----
				DD-UI	0.138	0.0345	0.710	-----
				FI-UI	0.477	0.0628	0.491	-----
SYN_HS_A	43.8	< .001	0.496	DD-FI	43.5	0.602	< .001	Yes
				DD-UI	0.853	0.379	< .001	Yes
				FI-UI	3.09	0.253	0.006	Yes
SYN_HS_B	5.09	0.079	0.169	DD-FI	5.02	0.204	0.025	Yes
				DD-UI	0.908	0.0885	0.341	-----
				FI-UI	1.65	0.117	0.199	-----
SYN_GD_A	7.49	0.024	0.205	DD-FI	0.000	0.002	0.984	-----
				DD-UI	5.27	0.213	0.022	Yes
				FI-UI	5.56	0.215	0.018	Yes
SYN_GD_B	0.926	0.629	0.0712	DD-FI	0.701	0.0764	0.402	-----
				DD-UI	0.00	0.00	1.00	-----
				FI-UI	0.701	0.0764	0.402	-----

3.2 Response time

3.2.1 Descriptive Findings

Figure 11 presents the average response times recorded across the three thematic mapping techniques for real-world and synthetic interpretation tasks. The results demonstrate notable differences in interpretation speed depending on both the spatial task and the cartographic representation employed.

For the real-world hotspot task (RW_HS), response times were relatively low across all three mapping techniques, indicating that participants quickly completed hotspot identification, DD produced the lowest response time, while FI and UI required slightly longer interpretation times.

In contrast, the real-world gradient task (RW_GD) produced substantially higher response time across all mapping techniques. This suggests that gradient interpretation required greater cognitive effort than hotspot identification. The differences between DD, FI, and UI in this task were relatively small, indicating that all three representations demanded similar levels of processing time for real-world gradient interpretation.

For the synthetic hotspot tasks, notable differences emerged among the mapping techniques, In SYN_HS_A, UI required differences longer response times compared to DD and FI, suggesting that participants may have needed additional time to interpret contour-based hotspot patterns in the synthetic environment. Similar tendencies were observed in SYN_HS_B, although the differences between techniques were less pronounced.

The synthetic gradient tasks (SYN_GD_A and SYN_GD_B) also revealed increased interpretation times, particularly for UI and FI. In SYN_GD_A, FI produced the highest response time, while in SYN_GD_B, UI showed the longest interpretation duration among all mapping techniques. These findings indicate that continuous and contour-based representations may require more detailed visual processing when interpreting gradual synthetic density transitions.

Overall, the response time analysis suggests that hotspot tasks were generally interpreted faster than gradient task across all mapping techniques. While DD representations frequently supported faster task completion, FI and UI often required longer interpretation times, particularly in synthetic gradient scenarios. The results therefore indicate that interpretation

efficiency varies depending on both spatial patterns being represented and the cartographic techniques used.

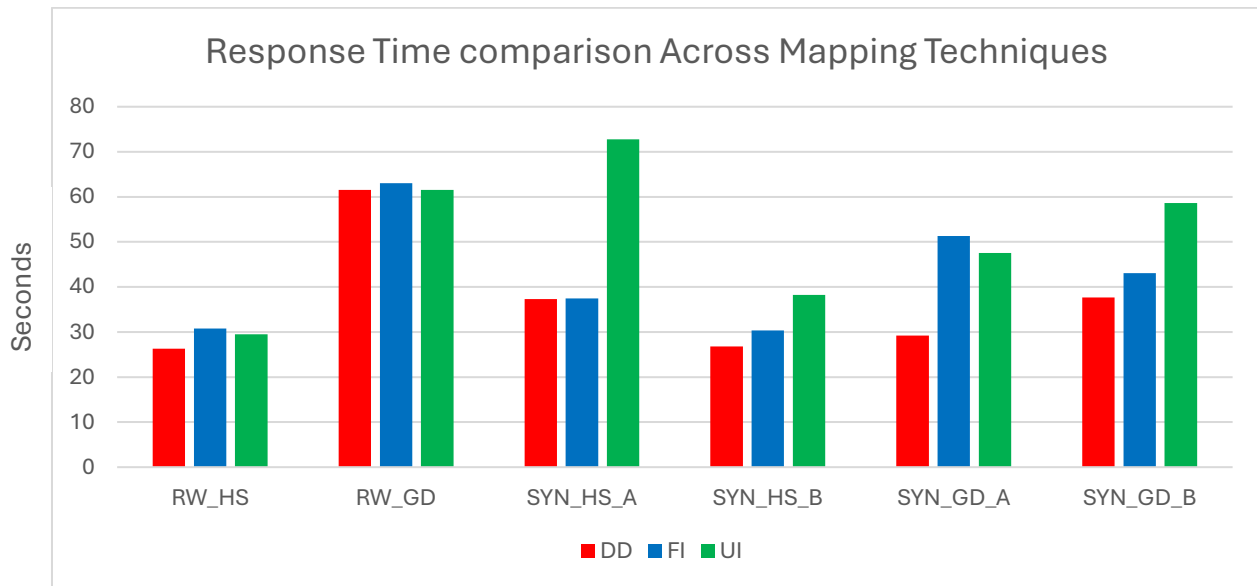


Figure 11: Average response time across thematic mapping techniques for real-world and synthetic spatial interpretation tasks

3.2.2 Statistical Significance

Response-time variables demonstrated positively skewed distribution and did not satisfy the assumptions of normality. Similarly, perceived difficulty ratings were measured using Likert-scale responses and showed non-normal distributions. Consequently, non-parametric Kruskal-Wallis tests were performed to compare differences in response time and perceived difficulty across the thematic mapping techniques.

Where statistically significant differences were identified between map types, post-hoc pairwise comparisons were conducted to determine which mapping techniques differed significantly from one another. Statistical significance was evaluated at the 0.05 significance level.

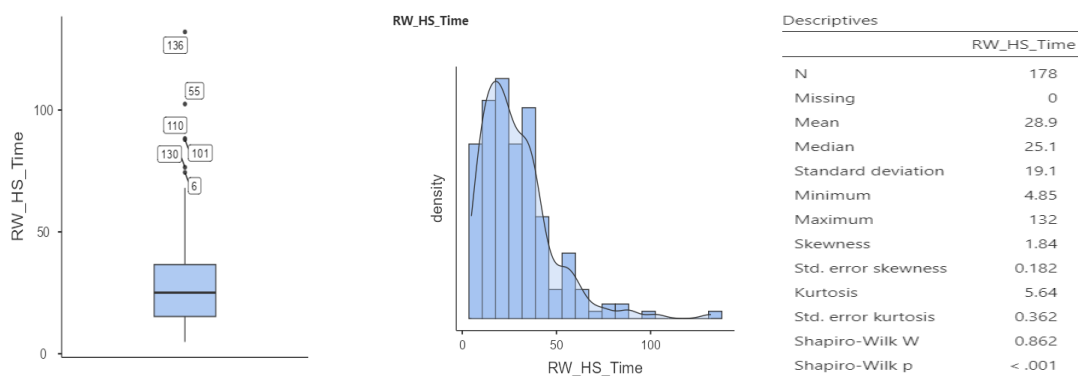


Figure 12: Normality Assessment Response Time outcome of one of the Variables

A Kruskal-Wallis's test was conducted to examine differences in response time across the three thematic mapping techniques. The analysis revealed a statistically significant difference in response time between map types, $X^2(2) = 8.23, p = 0.016$, with a small effect size ($e = 0.008$).

Post hoc Dunn's pairwise comparisons with Bonferroni correction showed a statistically significant difference between DD:FI maps ($p = 0.022$). However, no statistically significant differences were observed between FI:UI ($p = 1.000$) and DD:UI maps ($p = 0.077$). The test showed no statistically significant difference in response time between the three map types for the RW_HS task, $p = 0.309$. A statistically significant difference in response time between the three map types was revealed in RW_GD task ($p = 0.030$), the pairwise comparison are as follows FI:UI ($p = 0.033$), and there are not statistically significant between DD:FI $p = 1.000$ or DD:UI ($p = 0.165$).

For SYN_GD_A task $p = 0.002$, DD:FI ($p = 0.012$), DD:UI ($p = 0.005$). however, no statistically significant difference was observed between FI:UI ($p = 1.000$). there was not statistically significant between the map types for SYN_GD_B task, $p = 0.82$.

A statistically significant difference in response time between the three map types for SYN_HS_A was revealed $p = 0.036$, the pairwise comparison shows a statistically significant difference in FI:UI ($p = 0.031$). No statistically significant difference was observed in DD:FI ($p = 0.937$) or DD:UI ($p = 0.376$). for the SYN_HS_B task $p = 0.340$.

Table 4. Inferential Statistical analysis of response time across the three thematic map types.

Task	<i>Kruskal Wallis H</i>	<i>P</i>	Map Type	M (s)	SD	Post Hoc Groups	<i>p</i>
RW_HS	2.35	0.309	DD	26.3	15.8	-----	Not
			FI	30.72	17.9		Statistically
			UI	29.5	22.8		Significant
RW_GD	7.00	0.030	DD	61.5	46.0	-----	-----
			FI	63.0	39.3	-----	-----
			UI	61.5	81.3	FI-UI	0.033
SYN_GD_A	12.10	0.002	DD	29.2	14.6	DD-FI	0.012
			FI	51.3	77.9	DD-UI	0.005
			UI	47.5	33.7	-----	-----
SYN_GD_B	4.99	0.082	DD	37.7	25.8	-----	Not
			FI	43.0	21.2		Statistically
			UI	58.6	67.5		Significant
SYN_HS_A	6.67	0.036	DD	37.3	28.9	-----	-----
			FI	37.5	39.7	-----	-----
			UI	72.7	137	FI-UI	0.031
SYN_HS_B	2.16	0.340	DD	26.8	19.6	-----	Not
			FI	30.4	18.3		statistically
			UI	38.2	41.7		Significant

3.3 Perceived Difficulty Analysis

3.3.1 Descriptive Findings

The results shown in Figure 13 indicate that the average perceived difficulty ratings across the three thematic mapping techniques for both real-world and synthetic interpretation tasks. Overall, the results demonstrate noticeable differences in subjective task difficulty depending on both the cartographic representations and the spatial interpretation task. For the real-world hotspot task (RW_HS), DD received the lowest difficulty rating among the three mapping techniques, indicating that participants perceived hotspot identification using point-based

representations as relatively easy. In contrast, FI and UI produced slightly higher difficulty ratings for this task.

The real-world gradient task (RW_GD) generated higher perceived difficulty values across all mapping techniques compared to hotspot tasks. UI showed the highest difficulty rating, suggesting that contour-based representations may require greater cognitive effort when interpreting gradual density transitions. DD and FI demonstrated similar difficulty levels in this task.

For the synthetic hotspot tasks, FI consistently produced the lowest perceived difficulty values, particularly in SYN_HS_A. This suggests that continuous surface representation may support easier interpretation of synthetic clustered spatial patterns. In contrast, UI generally produced higher difficulty ratings, especially in SYN_HS_B. The synthetic gradient tasks (SYN_GD_A and SYN_GD_B) also revealed comparatively elevated difficulty ratings across all mapping techniques. UI produced the highest difficulty values in several gradient scenarios, while FI often demonstrated slightly lower perceived difficulty. These findings indicate that contour-based representations may increase interpretation complexity when participants attempt to identify gradual synthetic density variations.

Overall, the perceived difficulty analysis suggests that hotspot tasks were generally considered easier than gradient interpretation tasks across all mapping techniques. In addition, FI frequently demonstrated lower perceived difficulty in synthetic interpretation tasks, whereas UI often required greater cognitive effort. The results further highlight the relationship between cartographic representation, task complexity, and subjective user experience during spatial density interpretation.

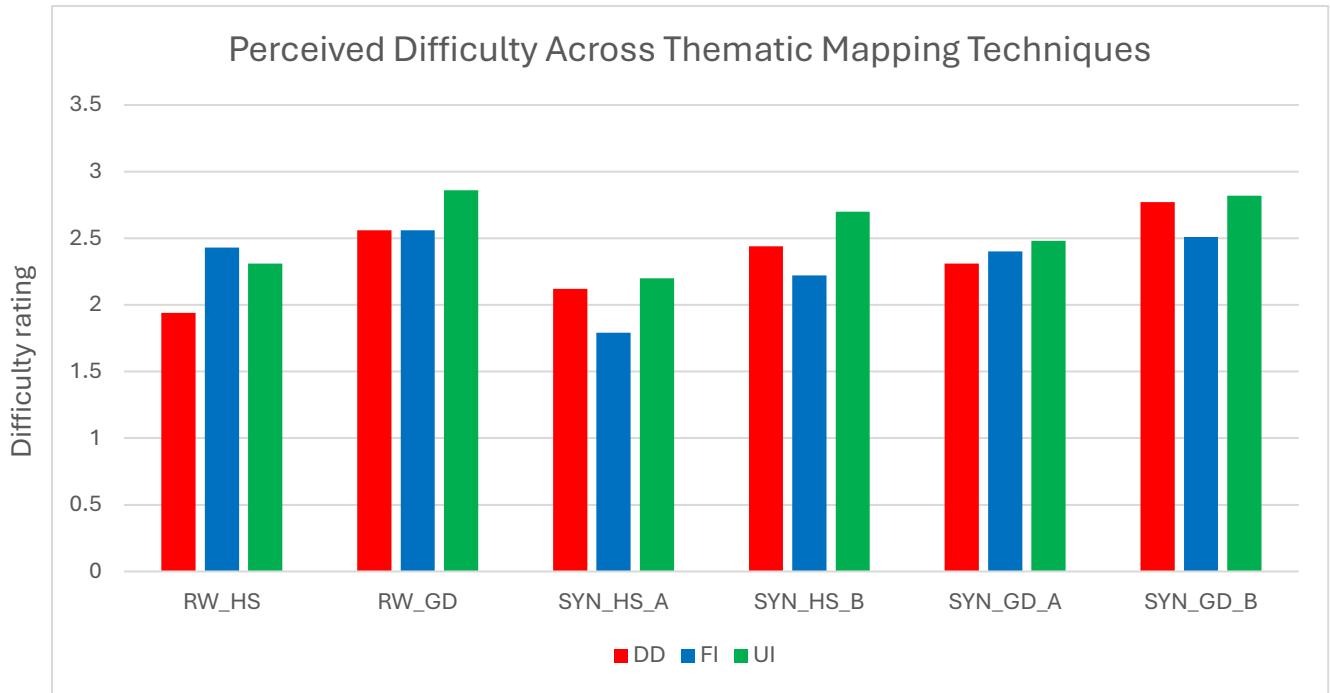


Figure 13: Average difficulty rating across thematic mapping techniques for real-world and synthetic spatial interpretation tasks

3.3.2 Statistical Significance

For the overall perceived task difficulty, A Kruskal-Wallis's test revealed a significant effect of the map on perceived task difficulty, $X^2(2) = 12.2$, $p = 0.002$. participants rated Filled Isopleth maps as the easiest ($M = 2.32$, Median = 2), followed by Dot Density maps ($M = 2.36$, Median = 2). Unfilled Isopleth received the highest difficulty ratings ($M = 2.57$, Median = 3). Dunn's post hoc comparisons with Bonferroni correction showed that Unfilled Isopleth maps were perceived as significantly more difficult than both Dot Density maps ($p = 0.030$) and Filled Isopleth maps ($p = 0.003$). No significant difference in perceived difficulty was observed between Dot Density and Filled Isopleth maps ($p = 1.000$).

RW_HS task shows a statistically significant difference in perceived difficulty among the three map types $p = 0.035$. However, only DD:FI $p = 0.026$ indicated a significant difference and does not exist between DD:UI $p = 0.226$ and FI:UI $p = 1.000$. No statistically significant difference among the three map types for the RW_GRD task $p = 0.218$. therefore, participants perceived the difficulty of the task similarly across DD, FI, and UI.

A significant difference in perceived difficulty was revealed in SYN_HS_A task, $p = 0.023$. FI:UI $p = 0.029$ and no significant difference were found between DD:FI $p = 0.112$ and DD:UI $p = 1.000$. Also, in SYN_HS_B, a significant difference was revealed among the three map

types $p = 0.027$. only FI:UI $p = 0.022$ shows significant difference and none exist between DD:FI $p = 0.684$ and DD:UI $p = 0.439$.

Perceived difficulty did not differ significantly among the three map types for the SYN_GRD_A task $p = 0.674$. consequently, not statistically significant pairwise differences were observed between any of the map types in the post hoc Dunn' tests (all Bonferroni-adjusted $p = 1.000$). Similarly, no significant difference in SYN_GRD_B task $p = 0.119$. none of the pairwise comparisons between map types were statistically significant (all Bonferroni adjusted $p > 0.05$).

Table 5. Inferential Statistical analysis of perceived difficulty rating across the three thematic map types.

Task	Kruskal-Wallis's X^2	p	Significant Pairwise Comparison	p
RW_HS	8.37	0.015	DD:FI	0.026
RW_GRD	3.05	0.218	-----	-----
SYN_HS_A	7.58	0.023	FI:UI	0.029
SYN_HS_B	7.20	0.027	FI:UI	0.022
SYN_GRD_A	0.79	0.674	-----	-----
SYN_GRD_B	4.25	0.119	-----	-----

3.4 Comparative Analysis of Mapping Techniques

3.4.1 Accuracy, Response Time, and Perceived Difficulty Trade-offs

The comparative analysis reveals important relationships between interpretation accuracy, response time, and perceived difficulty across the three thematic techniques. Overall, the results indicate that no single representation consistently outperformed the others across all spatial interpretation tasks. Instead, the effectiveness of each mapping technique depended strongly on the type of spatial pattern being interpreted.

Filled Isopleths (FI) frequently produced the highest accuracy values, particularly in gradient-related tasks and several synthetic scenarios. For example, FI clearly outperformed DD and UI in SYN_HS_A, SYN_HS_B, and both synthetic gradient tasks. However, these higher accuracy values were often associated with comparatively longer response times, especially in

SYN_GD_A and RW_GD. This suggests that participants may have required additional visual processing time to correctly interpret continuous density surfaces.

Despite the longer response times observed for FI in several tasks, perceived difficulty ratings for FI generally remained moderate and, in some synthetic tasks, lower than those of DD and UI. This may indicate that although FI representation required slightly longer interpretation times, participants still perceived them as cognitively manageable and visually supportive for interpreting gradual spatial transitions.

Dot Density (DD), on the other hand, frequently produced the shortest response times across multiple tasks, particularly in RW_HS, SYN_HS_B, SYN_GD_A and SYN_GD_B. This suggests that point-based representations may support faster visual scanning and quicker decision making. In addition, DD achieved the highest accuracy in the real-world hotspot task (RW_HS), reinforcing the idea that discrete point clustering supports hotspot localization effectively in realistic spatial environments. However, DD performance decreased considerably in synthetic hotspot and gradient tasks, particularly in SYN_HS_A and SYN_HS_B, where FI outperformed DD in accuracy. These findings suggest that while DD may support rapid interpretation, it may be less effective for representing continuous density transitions and controlled synthetic spatial structure.

Unfilled Isopleths (UI) generally demonstrated comparatively higher response times and higher perceived difficulty ratings across several tasks. UI produced the longest response times in SYN_HS_A and SYN_GD_B while also showing elevated perceived difficulty values in multiple gradient tasks. These patterns suggest that contour-based representations may impose greater cognitive demands on participants especially when interpreting complex synthetic gradients and clustered patterns. Although UI occasionally produced moderate accuracy values, its overall performance suggest that contour interpretation may require more intensive processing compared to DD and FI.

Taken together, the results indicate that important trade-offs between interpretation efficiency, accuracy, and subjective cognitive effort. Faster interpretations did not always correspond to higher accuracy, and representations associated with longer response times were not necessarily perceived as difficult. These findings highlight the complex relationship between cartographic representation, spatial task type, and human visual interpretation behaviour.

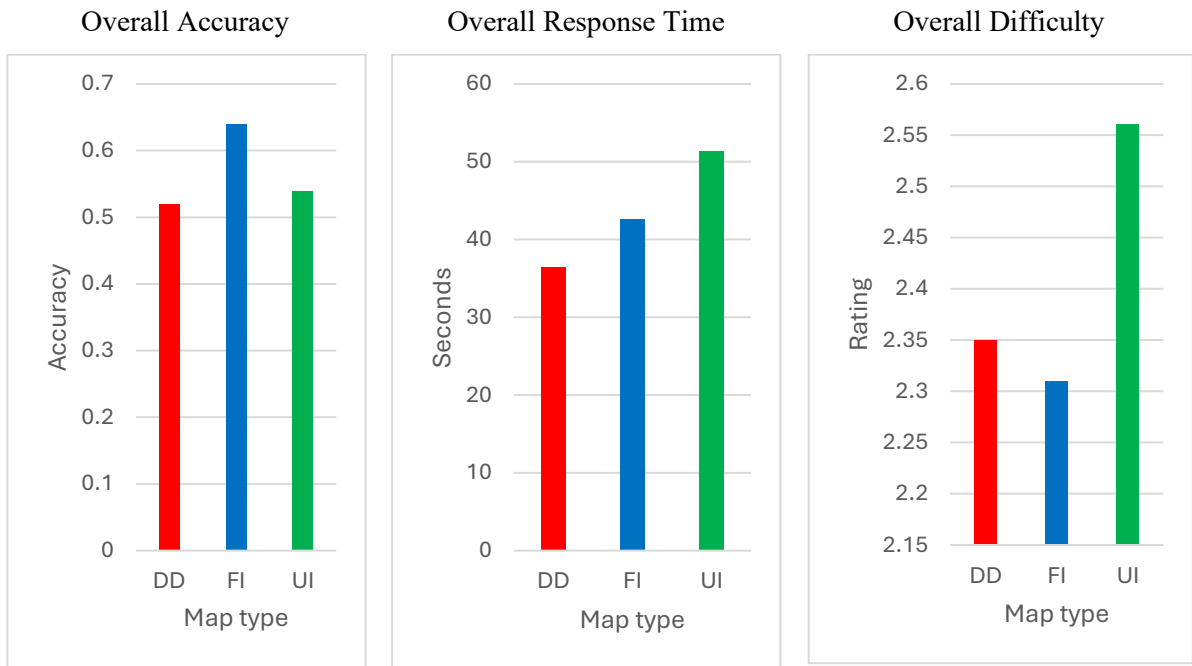


Figure 14: Comparative trade-offs between interpretation accuracy, response time, and perceived difficulty across thematic mapping techniques.

3.4.2 Hotspot VS Gradient Interpretation

3.4.2.1 Descriptive findings

The comparison between hotspot and gradient interpretation tasks revealed clear differences in participants performance and cognitive processing behaviour. Across all three thematic mapping techniques, hotspot tasks generally produced lower response times and lower perceived difficulty ratings than gradient interpretation tasks. This suggests that participants were able to identify concentrated spatial clusters more efficiently than gradual density transitions.

The real-world hotspot task (RW_HS) demonstrated relatively high accuracy and low response times across all mapping techniques, particularly for DD. This indicates that discrete point-based representations may support intuitive recognition of localized concentration patterns within realistic spatial environments. Participants also perceived hotspot identification tasks as comparatively easier than gradient interpretation tasks, further supporting the idea that clustered patterns are cognitively simpler to identify.

In contrast, gradient interpretation tasks consistently produced higher response times and higher perceived difficulty ratings. These trends were especially visible in RW_GD, SYN_GD_A, and SYN_GD_B, where participants required more time to interpret gradual spatial transitions. The increased cognitive demand associated with gradients likely results from the need to

continuously evaluate changing density levels across space rather than simply identifying concentrated clusters.

Filled Isopleths demonstrated particularly strong performance in gradient interpretation tasks. FI consistently achieved higher accuracy values in RW_GD, SYN_GD_A and SYN_GD_B, suggesting that continuous surface representations effectively communicate gradual spatial variation. In comparison, DD often showed lower accuracy in gradient scenarios, indicating that discrete point distributions may be less suitable for representing smooth density transitions.

These findings reinforce the idea that thematic mapping effectiveness is task dependent. While DD appears highly effective for hotspot localization, FI may provide superior support for gradient interpretation and continuous density analysis.

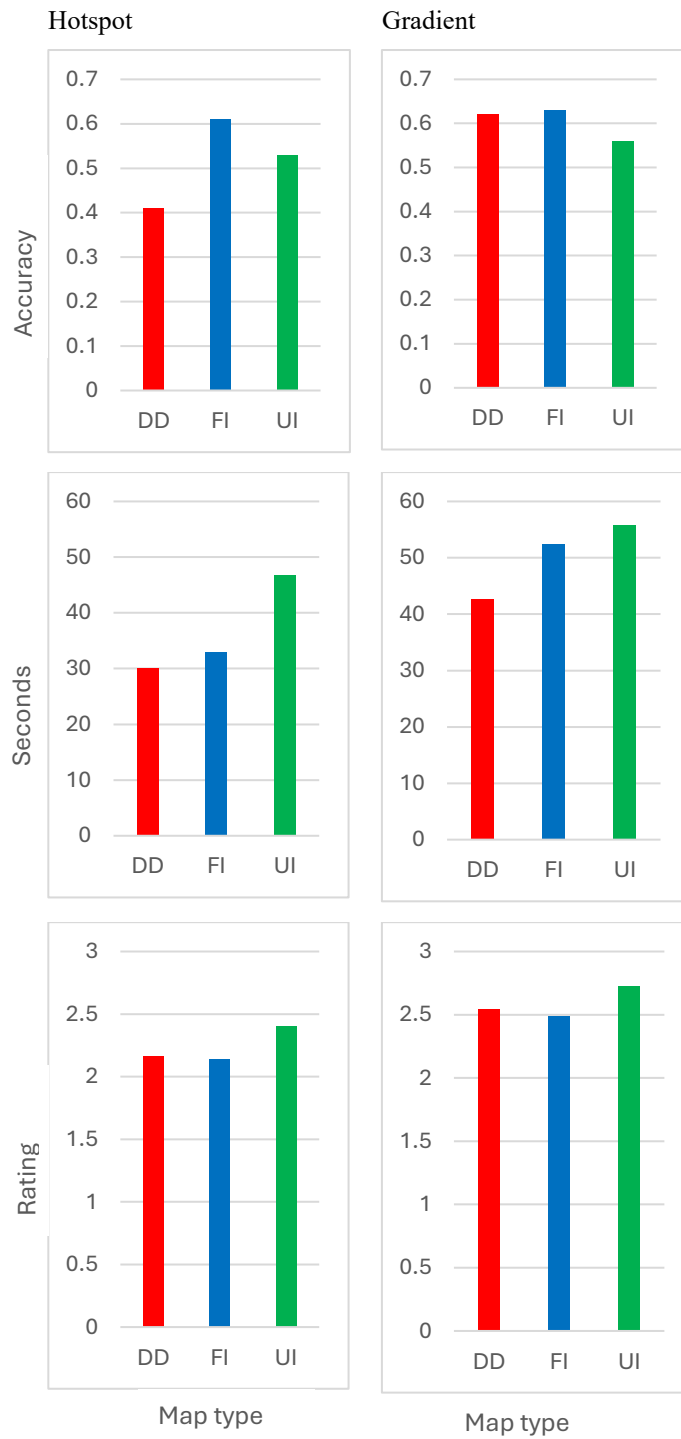


Figure 15: Comparative analysis of hotspot and gradient interpretation tasks across thematic mapping techniques. The figure summarizes differences in accuracy, response time, and perceived difficulty for DD, FI, and UI representation across real-world and synthetic spatial interpretation tasks.

3.4.2.2 Statistical Significance

For the hotspot identification tasks, answer accuracy differed significantly across the three thematic map types, $X^2(2, N = 534) = 14.3, p < 0.001$, Cramer's $V = 0.164$. FI achieved the highest accuracy (61.8%), followed by UI (53.4%) and DD (42.0%). These findings suggest that continuous filled surfaces supported hotspot identification more effectively than either contour-based or point-based representation.

In contrast, no statistically significant difference in answer accuracy was observed for gradient interpretation between the three thematic map types, $X^2(2, N = 534) = 4.98, p = 0.083$, Cramer's $V = 0.097$. Although FI produced the highest accuracy (68.3%), followed by DD (62.6%) and UI (56.9%), these differences were not sufficiently large to indicate a significant effect of map type on gradient interpretation performance.

Table 6: Chi-square test comparing answer accuracy across thematic map types for hotspot and identification and gradient interpretation tasks.

Spatial Task Type	X	<i>p</i> -value	Cramer's V
Hotspot	14.30	< 0.001	0.164
Gradient	4.98	0.083	0.097

3.4.3 Real-World VS Synthetic Tasks

The comparison between real-world and synthetic datasets also revealed notable differences in interpretation behaviour and mapping performance. Real-world tasks involved more irregular spatial distribution and greater visual complexity, whereas synthetic datasets provided controlled spatial structures designed to isolate specific density patterns.

In real-world tasks, performance differences between mapping techniques were often less pronounced. For example, although DD achieved the highest accuracy in RW_HS and FI performed best in RW_GD, the overall differences among techniques remained relatively moderate. This may reflect the influence of natural spatial variability and visual noise present in real-world population distributions.

In contrast, the synthetic tasks frequently produced stronger performance contrasts between mapping techniques. FI demonstrated particularly strong accuracy performance in SYN_HS_A and the synthetic gradient tasks, while DD showed substantially reduced accuracy in several synthetic scenarios. These stronger contrasts may result from the controlled nature of synthetic datasets, where spatial structures were intentionally simplified to emphasize specific hotspot and gradient characteristics.

Synthetic tasks also tended to produce larger differences in response time and perceived difficulty. UI demonstrated considerably higher response times and elevated perceived difficulty ratings in several synthetic tasks, suggesting that contour-based representations may become more cognitively demanding when participants attempt to interpret abstract or highly controlled spatial patterns.

Overall, the comparison between real-world and synthetic datasets demonstrates that dataset's structure and spatial complexity strongly influence thematic map interpretation. While real-world datasets introduce natural irregularities and contextual complexity, synthetic datasets provide clearer experimental control and reveal more pronounced differences in cartographic performance across mapping techniques.

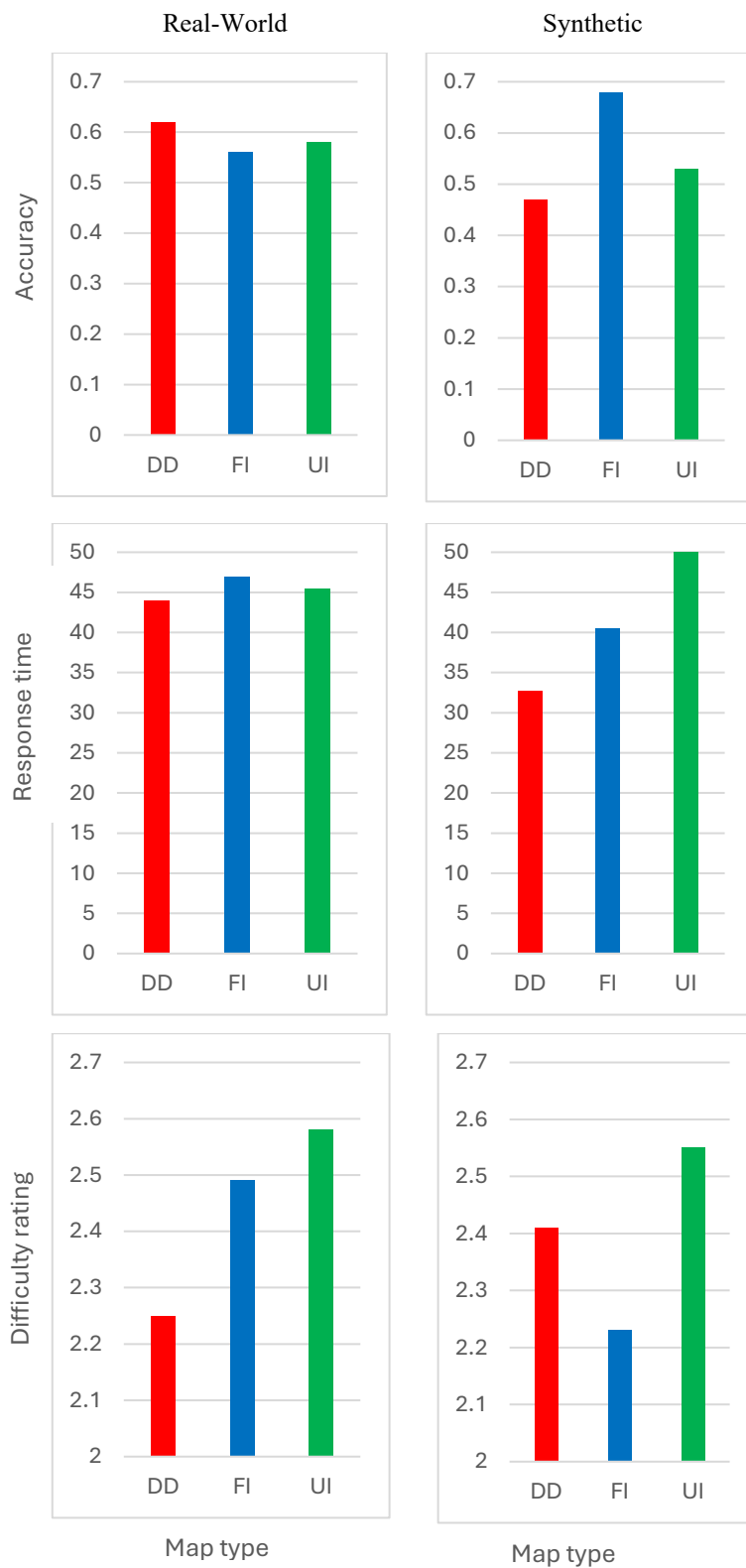


Figure 16: Comparison of interpretation performance between real-world and synthetic spatial tasks across thematic mapping techniques. Real-world tasks include RW_HS and RW_GD, while synthetic tasks include SYN_HS_A, SYN_HS_B, SYN_GD_A, and SYN_GD_B

3.5 Relationship Between Subjective Difficulty and Objective Performance

To investigate the relationship between subjective task difficulty and objective interpretation performance, perceived difficulty ratings were compared with measures of accuracy and response time. Because task difficulty was recorded using ordinal rating scales and accuracy was represented as a binary outcome (correct or incorrect), both correlation and group comparison approaches were employed. Spearman's rank correlation was used to examine the association between perceived difficulty and performance measures, while Mann-Whitney U tests were conducted to compare difficulty ratings between participants who answered correctly and those who answered incorrectly. Together, these analyses provide complementary insights into how subjective perceptions task difficulty relates to objective task performance.

3.5.1 Difficulty vs Accuracy

For the real-world hotspot task (RW_HS), higher perceived difficulty was associated with lower accuracy ($\rho = -0.155$, $p = 0.039$). participants who answered incorrectly also reported significantly higher difficulty ratings than those who answered correctly ($U = 2974$, $p = 0.040$). in contrast, real world gradient task (RW_GD), no significant relationship was observed between perceived difficulty and accuracy ($\rho = 0.044$, $p = 0.562$). similarly, difficulty ratings did not differ significantly between participants who answered correctly and those who answered incorrectly ($U = 3737$, $p = 0.561$).

For the synthetic hotspot task (SYN_HS_A), participants who perceived the task more difficult tended to achieve lower accuracy scores ($\rho = -0.226$, $p = 0.02$). This finding was further supported by significantly higher difficulty ratings among participants who answered incorrectly ($U = 2982$, $p = 0.03$) and in SYN_HS_B, perceived difficulty showed a weak negative association with accuracy ($\rho = -0.124$, $p = 0.100$). However, neither the relationship nor the difference in difficulty ratings between correct and incorrect responses was statistically significant ($U = 3372$, $p = 0.100$).

No significant relationship was observed in SYN_GD_A between perceived difficulty and accuracy ($\rho = -0.002$, $p = 0.976$). likewise, difficulty ratings did not differ significantly between participants who answered correctly and those who answered incorrectly ($U = 3559$, $p = 0.977$). Similarly, for SYN_GD_B, no significant relationship was observed between perceived difficulty and accuracy ($\rho = 0.001$, $p = 0.992$). also, difficulty ratings did not differ significantly between participants ($U = 3446$, $p = 0.994$).

Table 7: Results of correlation and group comparison analyses between perceived task difficulty and interpretation accuracy across experimental tasks

Task	Spearman ρ (Difficulty- Accuracy)	p	Mann-Whitney U	p
RW_HS	-0.155	0.039	2974	0.040
RW_GD	0.044	0.562	3737	0.561
SYN_HS_A	-0.226	0.02	2982	0.03
SYB_HS_B	-0.124	0.100	3372	0.100
SYN_GD_A	-0.002	0.976	3559	0.977
SYN_GD_B	0.001	0.992	3446	0.994

The relationship between perceived difficulty and accuracy was generally weak across the experimental tasks. Significant negative associations were observed for the real-world hotspot and synthetic hotspot task, indicating that participants perceived these tasks as more difficult tended to achieve lower scores. No significant relationship was found for the gradient interpretation tasks, where perceived difficulty appeared largely unrelated to performance. These findings suggest that subjective difficulty may have a greater influence on hotspot identification than on gradient interpretation tasks, although the observed effects were generally small.

3.5.2 Difficulty vs Response Time

For the real-world hotspot task, a weak positive relationship was observed between perceived difficulty and response time ($\rho = 0.015$, $p = 0.038$). A stronger positive relationship was found for the real-world gradient task ($\rho = 0.318$, $p < 0.001$), indicating that participants who perceived the task as more difficulty generally required more time to complete it. A similar pattern was observed for synthetic hotspot task A ($\rho = 0.217$, $p = 0.004$). Although positive correlations were also observed for synthetic hotspot task B ($\rho = 0.217$, $p = 0.072$), synthetic gradient task A ($\rho = 0.096$, $p = 0.204$), and synthetic gradient task B ($\rho = 0.133$, $p = 0.078$), these relationships were not statistically significant.

Table 8: Spearman correlation results between perceived task difficulty and response time across experimental tasks

Task	Spearman ρ (Difficulty-Response time)	p	Interpretation
RW_HS	0.156	0.038	Weak positive, significant
RW_GD	0.318	< 0.001	Moderate positive, significant
SYN_HS_A	0.217	0.004	Weak positive, significant
SYB_HS_B	0.135	0.072	Weak positive, not significant
SYN_GD_A	0.096	0.204	Very weak, not significant
SYN_GD_B	0.133	0.078	Weak, not significant

Overall, perceived difficulty was more consistently associated with response time than with accuracy. Significant positive relationships were observed for the real-world hotspot, and synthetic hotspot A, suggesting that participants tended to spend more time on tasks they perceived as difficult. In contrast, the relationship between perceived difficulty and accuracy was generally weak and only significant for two hotspots tasks. These findings suggest that subjective difficulty is more strongly reflected in task completion time than in interpretation accuracy.

3.6 Influence of Participants Characteristics

Participant characteristics are summarized in Table 1, Table 2, and Figure 6. Additional exploratory analyses were conducted to examine whether these characteristics influenced overall task performance. No statistically significant differences in overall accuracy were observed across educational levels (Table 1) or self-reported cartographic experience levels (Table 2), although the effect of cartographic experience approached significance. Furthermore, inspection of the age and gender distributions (Figure 6) revealed no substantial imbalances across the experimental groups. Overall, participant performance appeared to be largely independent of demographic and experience-related factors. Consequently, these variables were not considered further in the main analyses.

4. Discussion

4.1 Overall Performance of the Thematic Map Types

The results demonstrate that thematic map design significantly influenced participants' interpretation of population density patterns. Across all tasks, Filled Isopleth achieved the highest overall accuracy (65%), followed by Unfilled Isopleths (55%) and Dot Density maps (52%). At the same time, Dot Density maps generally produced the shortest response times, while unfilled Isopleths were frequently associated with the highest perceived difficulty ratings. These findings indicate that different thematic representations support spatial analysis tasks in different ways. Filled Isopleth appear to provide the strongest support for accurate interpretation. The higher accuracy achieved by Filled Isopleths may be attributed to their continuous representation of population density. Continuous colour variations allow users to identify areas of high and low density while simultaneously perceiving gradual spatial transitions Shaito et al., (2022). According to Zanin et al. (2013), the human visual system can rapidly distinguish between light and dark elements, making colour value an effective visual variable for communicating spatial structure and ordered pattern. As a result, Filled Isopleths may reduce the cognitive effort required to interpret density variations, allowing users to estimate spatial patterns more directly and accurately than with point-based or contour-based representations.

This aligns with an empirical user study by Słomska-Przech and Gołębiowska (2021), which confirmed that thematic map type significantly influenced interpretation performance and that map users achieved higher accuracy when using representations that effectively communicated overall spatial patterns. Their study further demonstrated that accuracy, response time, and perceived difficulty often reflected similar trends across map types.

Although Dot Density maps produced lower accuracy overall, they frequently allowed participants to respond more quickly, suggesting a reliance on rapid visual estimation of point concentration over detailed analytical processing (Roth et al., 2019, 2020). Faster responses therefore did not necessarily translate into more accurate interpretations. Taken together, these findings suggest that thematic map usability cannot be adequately described using a single performance indicator, as different representations may optimize several aspects of user performance.

4.2 Hotspot Identification Tasks

One of the objectives of this study was to compare the effectiveness of Dot Density, Filled Isopleths, and Unfilled Isopleths for hotspot identification. The result indicate that Filled

Isopleth generally supported the most accurate identification of population hotspots. The ability of Filled Isopleths to combine continuous colour gradients with immediate visual grouping likely reduced the cognitive effort required to recognize concentrated population regions, explaining their superior performance across hotspot identification. This finding is consistent with previous user-centred cartographic research showing that thematic map design influences map-reading performance and task success. Studies comparing alternative thematic representations have demonstrated that representations emphasizing overall spatial patterns tend to support more accurate interpretation than those requiring users to estimate values from discrete symbols or more complex visual structures (Słomska-Przech et al., 2021).

Dot Density maps also performed reasonably well for hotspot identification. The relatively good performance of Dot Density maps is unsurprising because dot-based representations are specifically designed to reveal local concentrations and spatial clustering. Previous user evaluations have shown that dot maps are often perceived as visually intuitive for recognizing the distribution of phenomena, although interpretation accuracy may decrease when users must estimate quantitative differences between areas rather than simply identify clusters (Wielebski, & Medyńska-Gulij, 2023). This decrement in precision is frequently driven by a certain visual discomfort caused by touching and overlapping dots, which remains a well-known flaw of this method when point densities become high.

Unfilled Isopleths produced lower accuracy and higher difficulty ratings in hotspot task. Unlike Filled Isopleths, raw contour lines lack explicit visual continuity, forcing users to manually track line spacing and mentally interpolate density values across unshaded fields. This additional cognitive effort likely contributed to a drop in user performance, validating the arguments of Peters and Meng (2015), who note that relying strictly on line layers without clear area-filled segmentation transforms map reading into “difficult endeavour”. Overall, the findings suggest that while all three map types can communicate hotspot locations, Filled Isopleths provide the clearest and most accessible representation for identifying areas of concentrated population density.

4.3 Spatial Gradient Interpretation

The differences between map types became even more apparent in gradient interpretation tasks. Compared with hotspot identification, gradient interpretation requires users to understand continuous changes in density across space and to identify transitions between areas of high and low population concentration.

Filled Isopleths consistently provided the strongest support for these tasks. According to Bertin’s theory of visual variables, colour value and lightness are uniquely effective for

communicating ordered quantitative information because they allow users to perceive magnitude differences almost automatically (as reviewed in Carpendale, 2003). Filled Isopleths exploit these visual properties by transforming population density into continuous visual field, enabling users to recognize gradual spatial transitions without extensive cognitive processing. This is further supported by Cebrykow (2025), who demonstrated that continuous visual representations of interpolated surfaces yield higher user interpretation performance than their discretized counterparts.

Gradient interpretation was more challenging with Dot Density maps, while areas of high and low density could often be identified, understanding gradual transitions required participants to estimate changes in point concentration across space. Such estimations introduce greater uncertainty and increase cognitive workload, particularly when gradients are subtle.

Unfilled interpretation was more challenging. Although contour lines explicitly indicate density levels, users must mentally reconstruct the continuous surface between contours. This additional interpretative step likely explains the lower accuracy and higher perceived difficulty observed in gradient-related tasks.

These findings are expected because gradient interpretation fundamentally involves understanding gradual changes rather than identifying clusters. Consequently, representations designed to communicate continuous variation naturally provide greater support for this type of task.

4.4 Real-World Versus Synthetic Datasets

The inclusion of both synthetic and real-world population datasets provided an opportunity to evaluate thematic map performance under different levels of spatial complexity.

Overall, participants tended to perform better on synthetic datasets than on real-world datasets. This outcome reflects a common challenge in cartographic perception research. Simplified experimental datasets are often designed to isolate specific spatial structures, whereas real-world datasets such as Ruhr region population contain noise, overlapping urban centres, and irregular geographic arrangements that increase interpretation difficulty. User studies have repeatedly shown that increasing visual and informational complexity can reduce map-reading performance and increase cognitive demands (Korycka – Skorupa & Gołębiowska, 2020).

However, the relative performance ranking of the three map types remained largely consistent across both dataset types. Filled Isopleths generally maintained the highest accuracy, while Dot Density maps continued to provide faster responses. The consistency observed across both dataset types strengthens confidence that the differences between map representations reflects genuine characteristics of the visualizations rather than properties of a particular dataset.

These findings strengthen the practical relevance of the study because they demonstrate that the observed advantages of Filled Isopleths extend beyond simplified experimental scenarios and remain evident when applied to realistic population distributions.

4.5 Response Time and Cognitive Efficiency

Response time provides important insight into the cognitive effort required to interpret different map types. Although accuracy represents effectiveness, response time reflects efficiency.

The results revealed an interesting trade-off between speed and accuracy. Dot Density maps frequently produced shorter response times, suggesting that participants could rapidly identify visual patterns using point distributions. However, this speed advantage did not consistently translate into higher accuracy. This pattern suggests that participants initially relied on the rapid visual detection of point clusters when using Dot Density maps. However, estimating precise density differences from individual points likely required additional visual judgement, reducing interpretation accuracy. In contrast Filled Isopleth required slightly more inspection time because participants processed continuous colour information, but this additional processing appears to have supported more reliable decisions,

Filled Isopleths, therefore, often required slightly longer inspection times but produced substantially higher accuracy. From a usability perspective, this trade-off may be beneficial because the modest increase in interpretation time resulted in substantially improved performance.

Unfilled Isopleths generally required the longest response times. This finding is consistent with their higher perceived difficulty ratings and lower accuracy scores. Participants likely needed additional time to interpret contour structures and estimate density values between lines.

This finding contrasts with the results reported in previous studies (Słomska-Przech and Gołębiowska, 2021; Sukraini et al., 2022), who found that choropleth maps achieved the highest accuracy, the shortest response times, and the lowest perceived difficulty among the tested thematic map types. In the present study no single representation dominated all usability measures.

4.6 Perceived Difficulty and Objective Performance

Another important objective of this study was to assess perceived cognitive difficulty and examine its relationship with objective performance measures.

A clear relationship emerged between subjective difficulty ratings and objective task performance. Tasks that participants rated as more difficult generally exhibited lower accuracy and longer times.

This result is particularly noteworthy because several previous cartographic studies reported discrepancies between subjective and objective measures. For example, users sometimes preferred familiar map types even when those representations did not produce the best performance outcomes. Similar relationships between evaluations and objective measures have been reported in thematic map usability studies. Słomska-Przech and Gołębiowska (2021) found that map types producing higher accuracy frequently received more favourable usability assessments, suggesting that users are often capable of recognizing representations that support effective task completion.

In this study, Filled Isopleths were generally associated with lower perceived difficulty, higher accuracy, and favourable response times. This convergence of subjective and objective measures suggests that participants correctly perceived the relative ease of interpreting different map types. This correspondence between subjective and objective measures indicates that participants were generally able to accurately assess the cognitive demands imposed by each thematic representation.

The findings therefore support the notion that cognitive workload plays a meaningful role in thematic map interpretation. Representations that reduce the need for mental interpolation and visual estimation appear to facilitate both better performance and more positive user experiences.

4.7 Influence of Participants Characteristics

Exploratory analyses indicated that participants characteristics, including age group, gender, educational level and self-reported cartographic experience, did not significantly influence overall task accuracy. The absence of strong demographic effects is notable because previous research frequently documents significant disparities in map-reading performance tied to individual user characteristics such as spatial literacy, gender, and prior experience (Havelková & Hanus, 2019). One possible explanation for this study's deviation is the experimental configuration. Before beginning the survey, participants received detailed instructions, explanatory examples, and practice material illustrating how each map type should be interpreted.

These introductory materials likely reduced performance gaps between participants by establishing a common baseline level of understanding, allowing those with limited cartographic experience to perform similarly to those with greater experience.

This interpretation aligns with the cognitive framework reviewed by Pan and Carpenter (2023), who synthesized how initial exposures, pre-tests, and introductory prompts serve as powerful mechanisms that prime cognitive structures and successfully minimize baseline variances

among learners. Therefore, the results suggest that the observed differences in performance were primarily driven by the map representation themselves rather than by participants characteristics.

4.8 Implications for Cartographic Design

The findings of this study have several implications for thematic map design. First, Filled Isopleths appear particularly well suited for communicating continuous population density patterns and supporting both hotspot identification and gradient interpretation. Their ability to combine high accuracy with relatively low perceived difficulty makes them an effective choice for general audiences.

Second, Dot Density maps remain valuable when the objective is to communicate local clustering and provide an intuitive representation of population distribution. Although accuracy was lower than for Filled Isopleths, their rapid interpretation times suggest that they remain useful for exploratory visualization.

Finally, Unfilled Isopleths appear less suitable for non-expert audiences when accurate interpretation of population density is required. The higher cognitive demands associated with contour resulted in lower performance across several tasks.

Taken together, the results support a growing body of cartographic usability research emphasizing that thematic map design should be matched to the intended analytical task. Consistent with previous user studies (Korycka–Skorupa & Gołębiowska, 2020; Słomska-Przech & Gołębiowska, 2021; Shaito et al., 2022), the present study demonstrates that thematic map choice has a substantial influence on spatial perception and interpretation. The results therefore support the argument that cartographic design decisions should be guided not only by aesthetic considerations but also by the analytical tasks users are expected to perform.

Overall, the findings demonstrate that selecting an appropriate thematic representation should depend on the intended analytical task rather than assuming that a single map type is universally optimal. This reinforces the importance of evidence-based cartographic design, where map selection is guided by empirical evaluations of user rather than convention alone.

4.9 Study Limitation

Although this study provides valuable insight into how different thematic map representations influence the interpretation of population density patterns, several limitations should be acknowledged.

First, the experiment was conducted through an online survey, meaning that participants completed the tasks using their own devices and in different environments. As a result, factors

such as screen size, display quality, lighting conditions, and potential distractions could not be controlled and may have influenced individual performance.

Second, this study focused specifically on three thematic mapping techniques; Dot Density maps, Filled Isopleths, and Unfilled Isopleths for representing population density. Therefore, the findings should not be generalized to other thematic map types or different geographic variables without further investigation.

Finally, although the use of both synthetic and real-world-datasets strengthened the study by allowing controlled and realistic evaluations, the synthetic datasets were intentionally simplified to highlight specific hotspot and gradient patterns. Consequently, they cannot fully represent the complexity of real-world geographic phenomena.

Despite these limitations, the combination of controlled experimental design, multiple usability measures, and the use of both synthetic and real-world datasets provides confidence that the findings offer meaningful contributions to cartographic perception research and practical guidance for thematic map design.

5. CONCLUSION

The conducted user study showed that thematic map type influence how users interpret spatial population density patterns. Filled Isopleths generally produced the highest accuracy and lowest perceived difficulty, while Dot Density maps often enabled faster responses. Unfilled Isopleths were generally associated with higher difficulty and lower performance across several tasks.

The results also demonstrated that map effectiveness depends on the task being performed. Filled Isopleth were particularly effective for communicating continuous density patterns and supporting gradient interpretation, whereas Dot Density maps remained useful for identifying local concentrations and clustering patterns. The inclusion of both synthetic and real-world datasets showed that these patterns were largely consistent across different spatial contexts.

Although the collected data generally favoured Filled Isopleth, the findings do not suggest that a single map type should be used in all situations. Each representation offers specific advantages depending on the purpose of the map and the information being communicated. Therefore, the selection of thematic mapping technique should be guided by the analytical task, the characteristics of the data, and the needs of the intended audience.

References

- [1] Ashour, L. (2024). A review of user-friendly freely-available statistical analysis software for medical researchers and biostatisticians. *Research in Statistics*, 2(1). <https://doi.org/10.1080/27684520.2024.2322630>
- [2] Beitlova, M., Popelka, S., & Vozenilek, V. (2020). Differences in Thematic Map Reading by Students and Their Geography Teacher. *ISPRS International Journal of Geo-Information*, 9(9), 492. <https://doi.org/10.3390/ijgi9090492>
- [3] Bertin, J. (1983). *Semiology of graphics; diagrams networks maps*. (University of Wisconsin, Press, Madison,
- [4] Besançon, L., Cooper, M., Ynnerman, A., & Vernier, F. (2020). *An evaluation of visualization methods for population statistics based on choropleths maps*. arXiv. <https://arxiv.org/abs/2005.00324>
- [5] Bondarenko M., Priyatikanto R., Tejedor-Garavito N., Zhang W., McKeen T., Cunningham A., Woods T., Hilton J., Cihan D., Nosatiuk B., Brinkhoff T., Tatem A., Sorichetta A., *Constrained estimates of 2015-2030 total number of people per grid square at a resolution of 3 arc (approximately 100m at the equator) R2025A version v1. Global Demographic Data project – Funded by The Bill and Melinda Gates Foundation (INV-045237). WorldPop – School of Geography and Environmental Science, University of Southampton*. Doi: 10:5258/SOTON/WP00839
- [6] Carpendale, M.S.T. (2003). *Considering Visual Variables as a Basis for Information Visualisation* (Technical Report 2003-734-37). Department of Computer Science, University of Calgary.
- [7] Cebrykow, M. (2025). Using and isopleth map to determine the population of any area. *Journal of Geography, Politics and Society*. 15(1), 106-115
- [8] Çöltekin, A., Heil, B., Garlandini, S., & Fabrikant, S. I. (2009). Evaluating the Effectiveness of Interactive Map Interface Designs: A Case Study Integrating Usability Metrics with Eye-Movement Analysis. *Cartography and Geographic Information Science*, 36(1), 5–17. <https://doi.org/10.1559/152304009787340197>
- [9] Crump MJC, McDonnell JV, Gureckis TM (2013) Evaluating Amazon's Mechanical Turk as a Tool for Experimental Behavioural Research. *PLOS ONE* 8(3): e57410. <https://doi.org/10.1371/journal.pone.0057410>

- [10] Ding, L., & Meng, L. (2014). A comparative study of thematic mapping and scientific visualization. *Annals of GIS*, 20(1), 23–37. <https://doi.org/10.1080/19475683.2013.862856>
- [11] Ehrlich, D., Kemper, T., Pesaresi, M., & Corbane, C. (2018). Built-up area and population density: Two essential societal variables to address climate hazard impact. *Environmental Science & Policy*, 90, 73-82. <https://doi.org/10.1016/j.envsci.2018.10.001>
- [12] Forest, D., & Medyńska-Gulij, B (2021). *Which mapping technique for population density is effective, attractive, and suggestive?* Abstracts of the International Cartographic Association, 3, 1. <https://doi.org/10.5194/ica-abs-3-82-2021>
- [13] GDAM. (2024). Database of Global Administrative Areas (Version 4.1). Retrieved October 12, 2025, from <https://gadm.org>
- [14] Havelková, L., & Hanus, M. (2019). Map Skills in Education: A Systematic Review of Terminology, Methodology, and Influencing Factors. *Review of International Geographical Education Online*, 9(2), 361-401. <https://doi.org/10.33403/rigeo.591094>
- [15] Korycka-Skorupa, J., & Gołębiowska, I. M. (2020). Numbers on Thematic Maps: Helpful Simplicity or Too Raw to Be Useful for Map Reading? *ISPRS International Journal of Geo-Information*, 9(7), 415. <https://doi.org/10.3390/ijgi9070415>
- [16] Lazar, J., Feng, J. H., & Hochheiser, H. (2017). Research methods in human-computer interaction (2nd ed.). Morgan Kaufmann Publishers.
- [17] M. He, X. Tang, and Y. Huang, "To visualize spatial data using thematic maps combined with infographics," *2011 19th International Conference on Geoinformatics*, Shanghai, China, 2011, pp. 1-5, Doi: 10.1109/GeoInformatics.2011.5980880.
- [18] MacEachren, A.M. (2004). *How maps work: Representation, Visualization, and design* (2nd ed.). Guilford Press
- [19] Medyńska-Gulij, B., Wielebski, Ł., Halik, Ł., & Smaczyński, M. (2020). Complexity Level of People Gathering Presentation on an Animated Map—Objective Effectiveness Versus Expert Opinion. *ISPRS International Journal of Geo-Information*, 9(2), 117. <https://doi.org/10.3390/ijgi9020117>
- [20] Pan, S.C., Carpenter, S.K. Prequestioning and Pretesting Effects: A Review of Empirical Research, Theoretical Perspectives, and Implications for Educational

- Practice. *Educ Psychol Rev* **35**, 97 (2023). <https://doi.org/10.1007/s10648-023-09814-5>
- [21] Peer, E., Brandimarte, L., Samat, S., & Acquisti, A. (2017). Beyond the Turk: Alternative platforms for crowdsourcing behavioural research. *Journal of Experimental Social Psychology*, 70, 153–163. <https://doi.org/10.1016/j.jesp.2017.01.006>
- [22] Peters, S. and Meng, L., 2015. Spatial-temporal density mapping for spatially extended dynamic phenomena – a novel approach to incorporate movements in density maps. *International Journal on Advances in Intelligent Systems*, 8(1 & 2), pp.27-39.
- [23] Pickle, L. W. (2003). Usability testing of map designs. In Proceedings of the 22nd International Cartographic Conference (ICC 2003). International Cartographic Association.
- [24] R core Team (2025). R: *A Language and environment for statistical computing*. (Version 4.5) [Computer software]. Retrieved from <https://cran.r-project.org>. (R packages retrieved from CRAN snapshot 2025-05-25)
- [25] Roth, R. E. (2013). Interactive maps: What we know and what we need to know. *Journal of Spatial Information Science*, (6), 59–115. <https://doi.org/10.5311/JOSIS.2013.6.105>
- [26] Roth, R. E., Çöltekin, A., Delazari, L., Filho, H. F., Griffin, A., Hall, A., ... van Elzakker, C. P. J. M. (2017). User studies in cartography: opportunities for empirical research on interactive maps and visualizations. *International Journal of Cartography*, 3(sup1), 61–89. <https://doi.org/10.1080/23729333.2017.1288534>
- [27] Roth, R. E., Kelly, M., Underwood, N., Vincent, K., & Sack, C. (2019). Interactive & Multiscale Thematic Maps: A preliminary Study. *Abstracts of the ICA*, 1, 315. <https://doi.org/10.5194/ica-abs-1-315-2019>
- [28] Roth, R.E., Kelly, M., Underwood, N., Lally, N., Liu, X., Vincent, K., & Sack, C. (2020). Interactive & Multiscale Thematic Maps: Preliminary Results from an Empirical Study. *Abstracts of the ICA*, 2, 135. <https://doi.org/10.5194/ica-abs-2-135-2020>
- [29] Şahin, M., & Aybek, E. (2020). Jamovi: An Easy-to-Use Statistical Software for the Social Scientists. *International Journal of Assessment Tools in Education*, 6(4), 670-692. <https://doi.org/10.21449/ijate.661803>

- [30] Shaito M, Elmasri R et al., Comparison of Map Visualization Techniques Used for Spatial and Spatio-Temporal Data: Analytical Survey Applied to Covid-19 Data, *Medical Research Archives*, [online] 10(9) <https://doi.org/10.18103/mra.v10i9.3072>
- [31] Silva de Souza, R., Sequeira, C. A., & Borges, E. M. (2024). Enhancing statistical education in chemistry and STEAM using JAMOVİ. Part 1: Descriptive statistics and comparing independent groups. *Journal of Chemical Education*, 101(12), 5027–5039.
- [32] Slocum, T.A., McMaster, R.B., Kessler, F. C., & Howard, H.H. (2023). *Thematic cartography and geovisualization* (5th ed.) CRC Press. <https://doi.org/10.1201/9781003150527>
- [33] Słomska-Przech, K., & Gołębiowska, I. M. (2021). Do Different Map Types Support Map Reading Equally? Comparing Choropleth, Graduated Symbols, and Isoline Maps for Map Use Tasks. *ISPRS International Journal of Geo-Information*, 10(2), 69. <https://doi.org/10.3390/ijgi10020069>
- [34] Sukraini, T. T., Yasa, I. M. A., & Wiguna, P. P. K. (2022). Comparing choropleth and graduated symbols: How different map types affect public understanding in COVID-19 map reading in Badung Regency, Bali, Indonesia. *Geographia Technica*, 17(1), 150–166. https://doi.org/10.21163/GT_2022.171.12
- [35] The jamovi project (2025). Jamovi. (Version 2.7) [Computer Software]. Retrieved from <https://www.jamovi.org>
- [36] Traun, C., Klug, H., & Burkhard, B. (2017). Mapping techniques. In B. Burkhard & J. Maes (Eds.), *Mapping ecosystem services* (pp. 55-62). Pensoft Publishers
- [37] Traun, C., Schreyer, M. L., & Wallentin, G. (2021). Empirical Insights from a Study on Outlier Preserving Value Generalization in Animated Choropleth Maps. *ISPRS International Journal of Geo-Information*, 10(4), 208. <https://doi.org/10.3390/ijgi10040208>
- [38] Wielebski, Ł., & Medyńska-Gulij, B. (2023). User Evaluation of Thematic Maps on Operational Areas of Rescue Helicopters. *ISPRS International Journal of Geo-Information*, 12(2), 30. <https://doi.org/10.3390/ijgi12020030>

- [39] Zanin, C., Lambert, N., Davoine, P. -A., Mathan, H., & Saint-Marc, C. (2013). Mapping guide. ESPON – Inspire Policy Making with Territorial Evidence. <https://hal.science/hal-03594611>

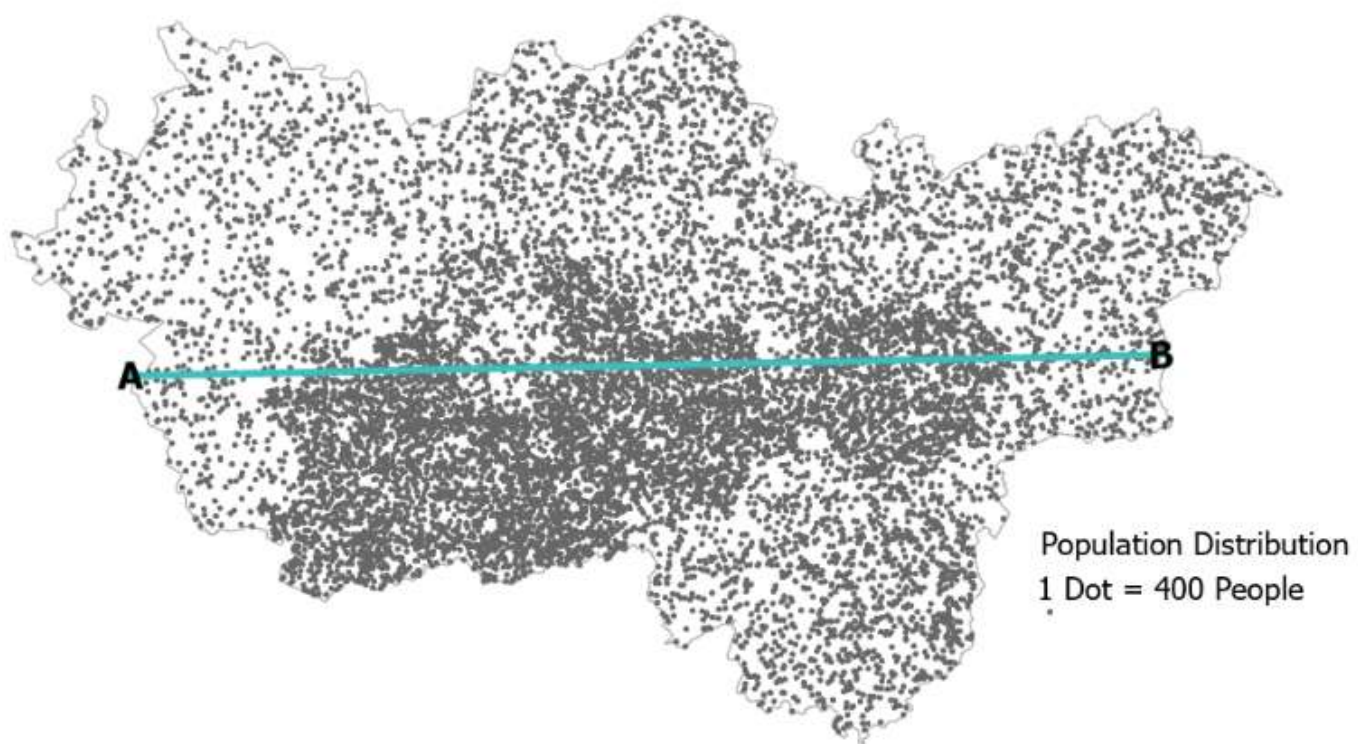
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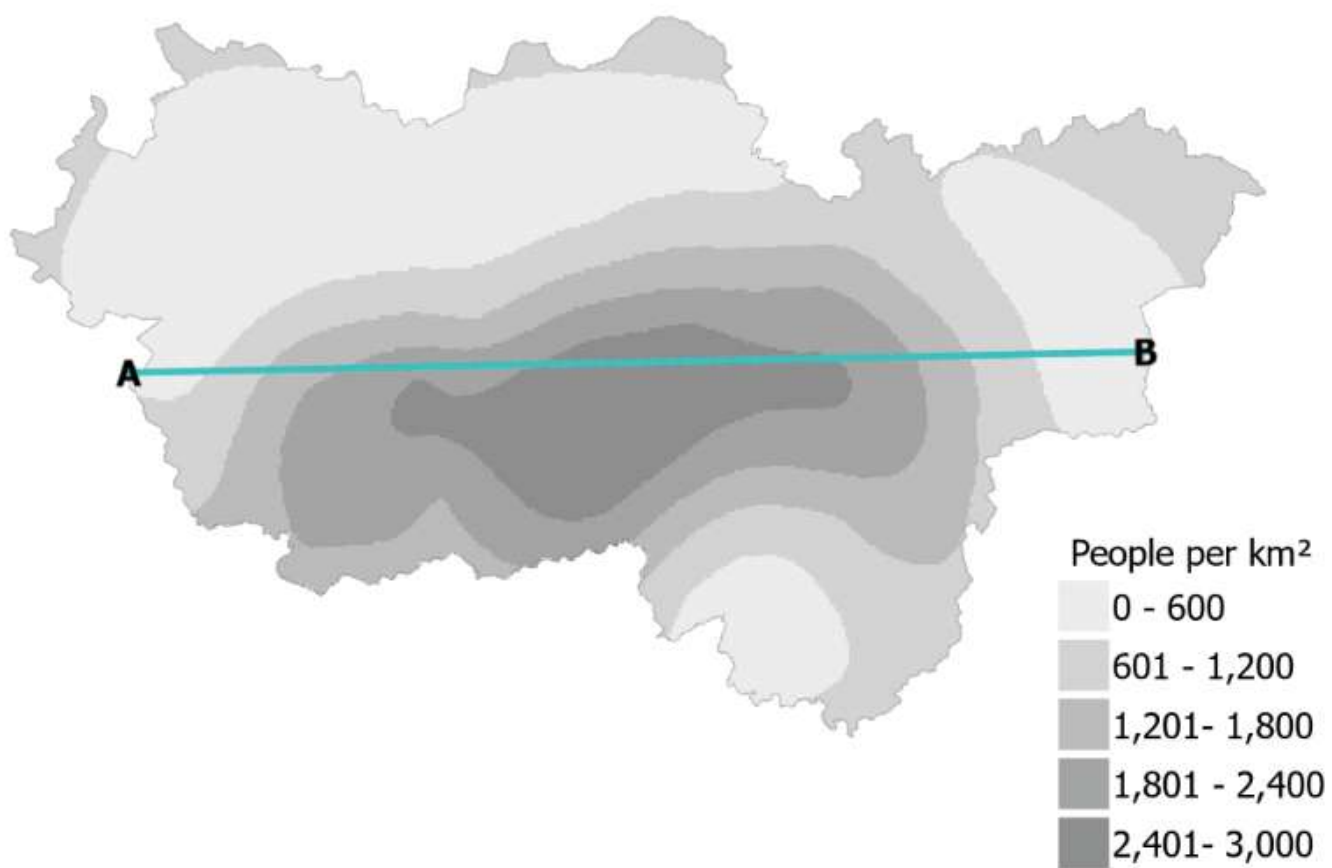
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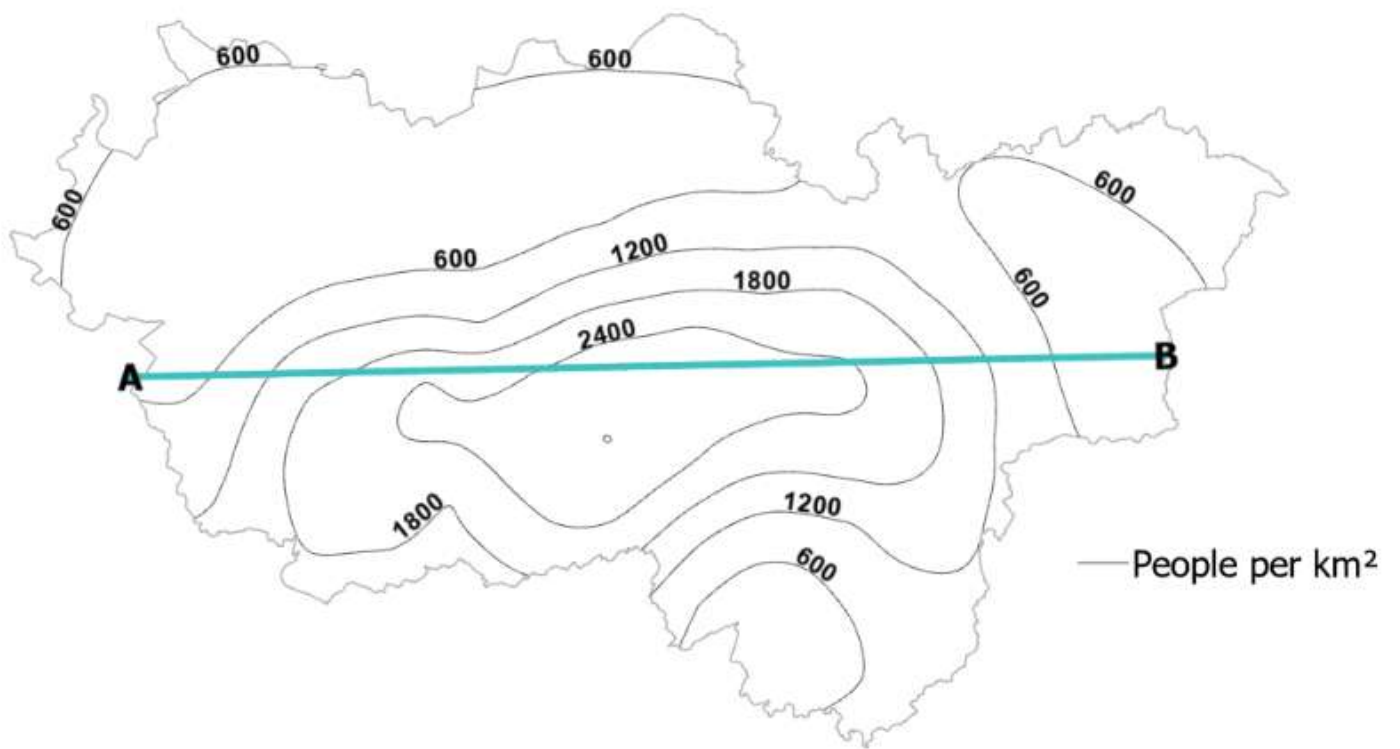
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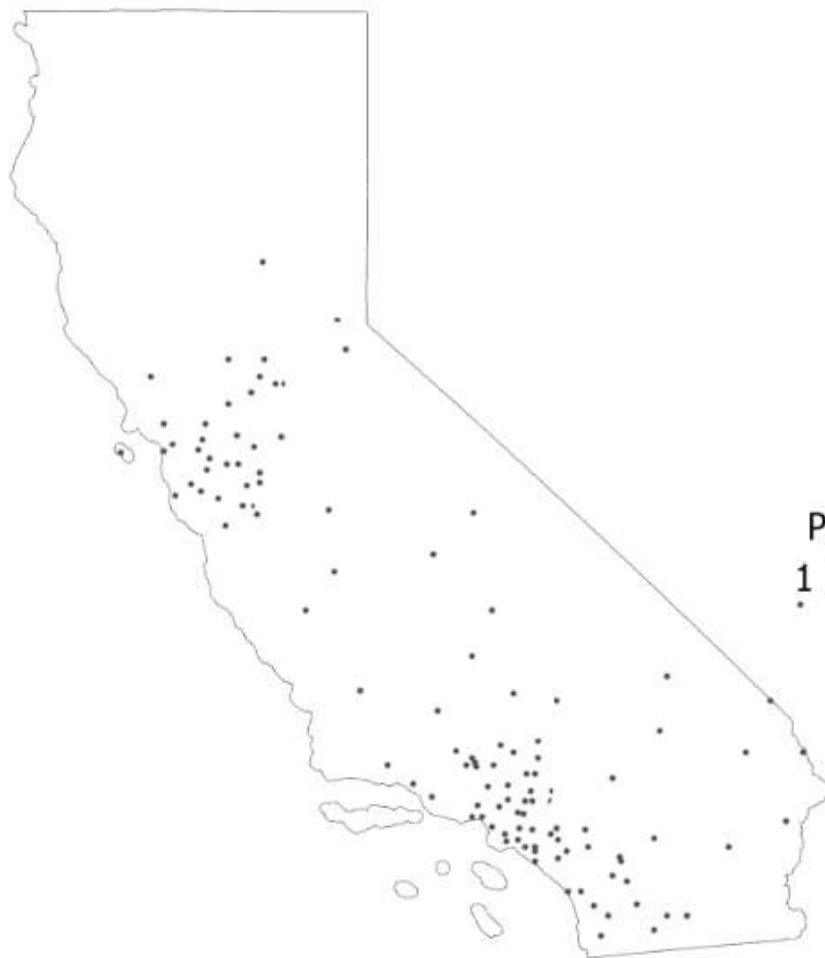
RW_GD



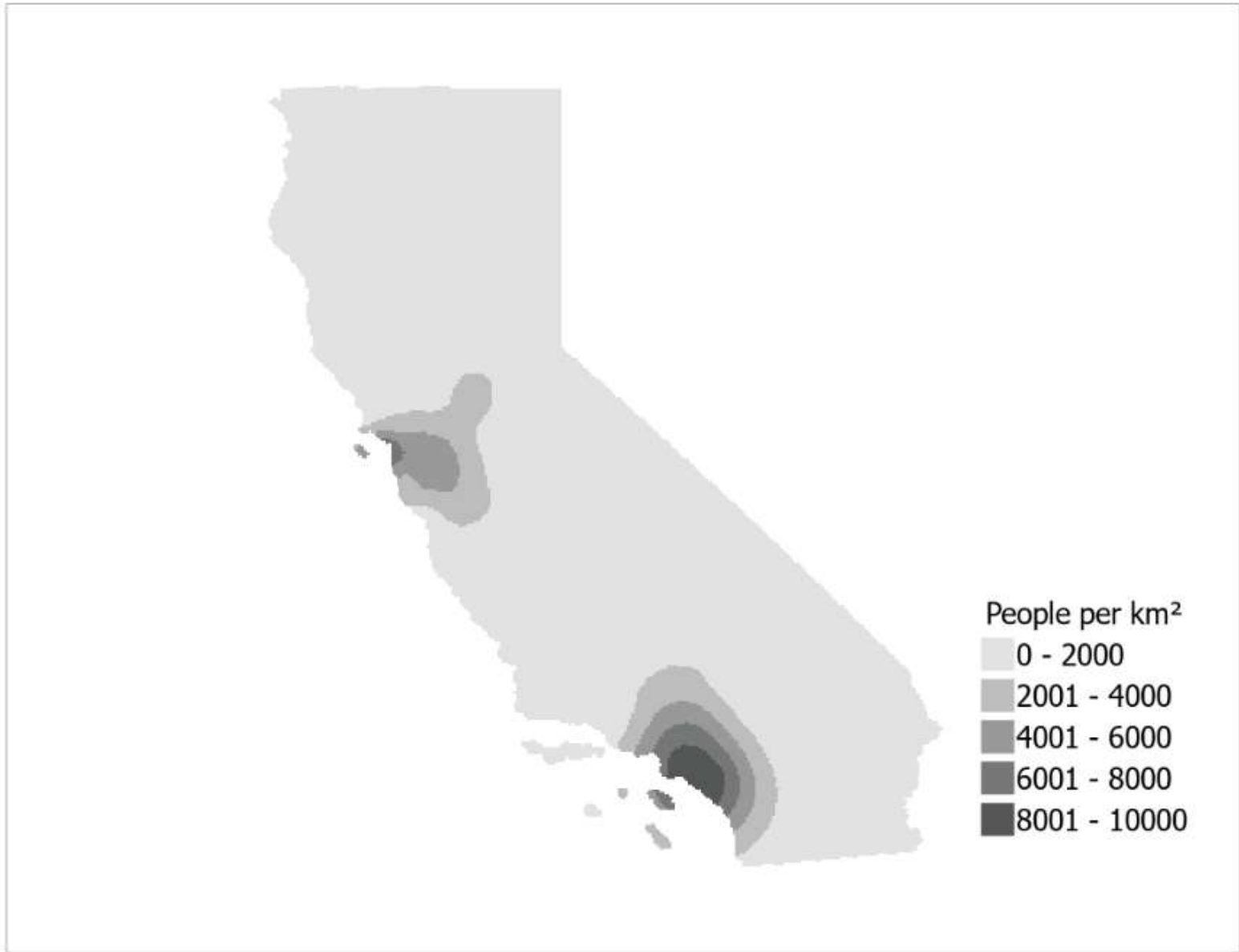




RW_HS

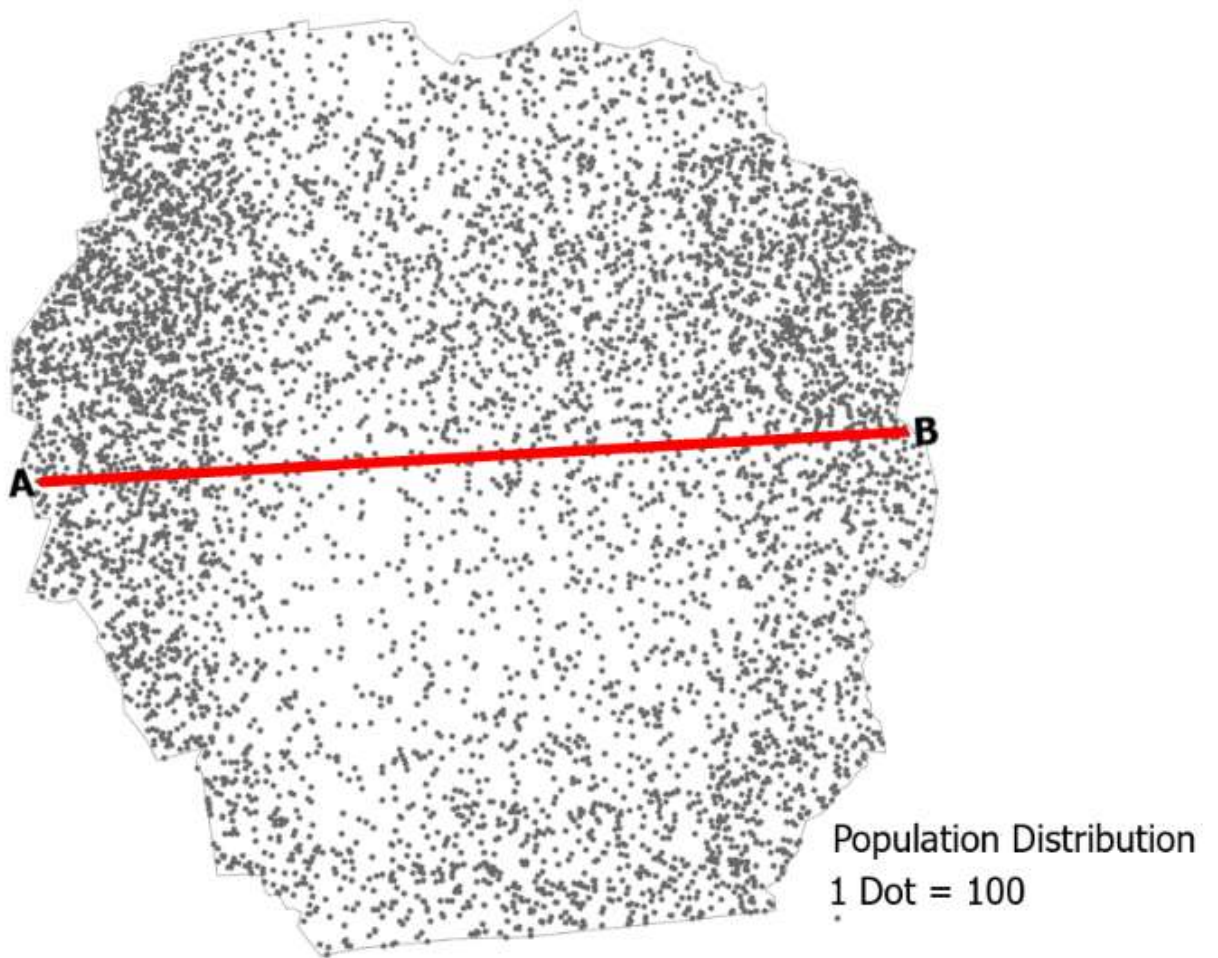


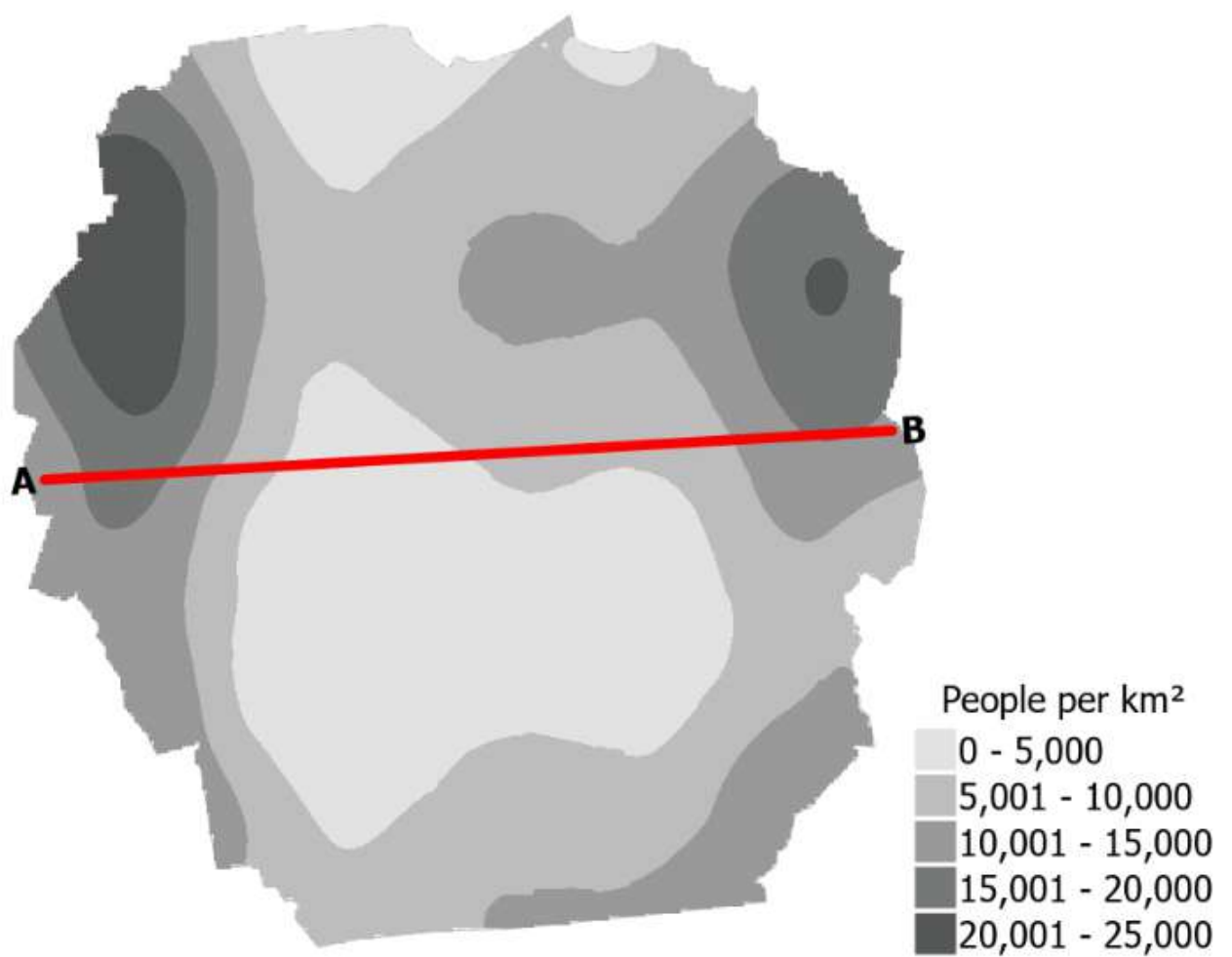
Population Distribution
1 Dot = 350,000 People

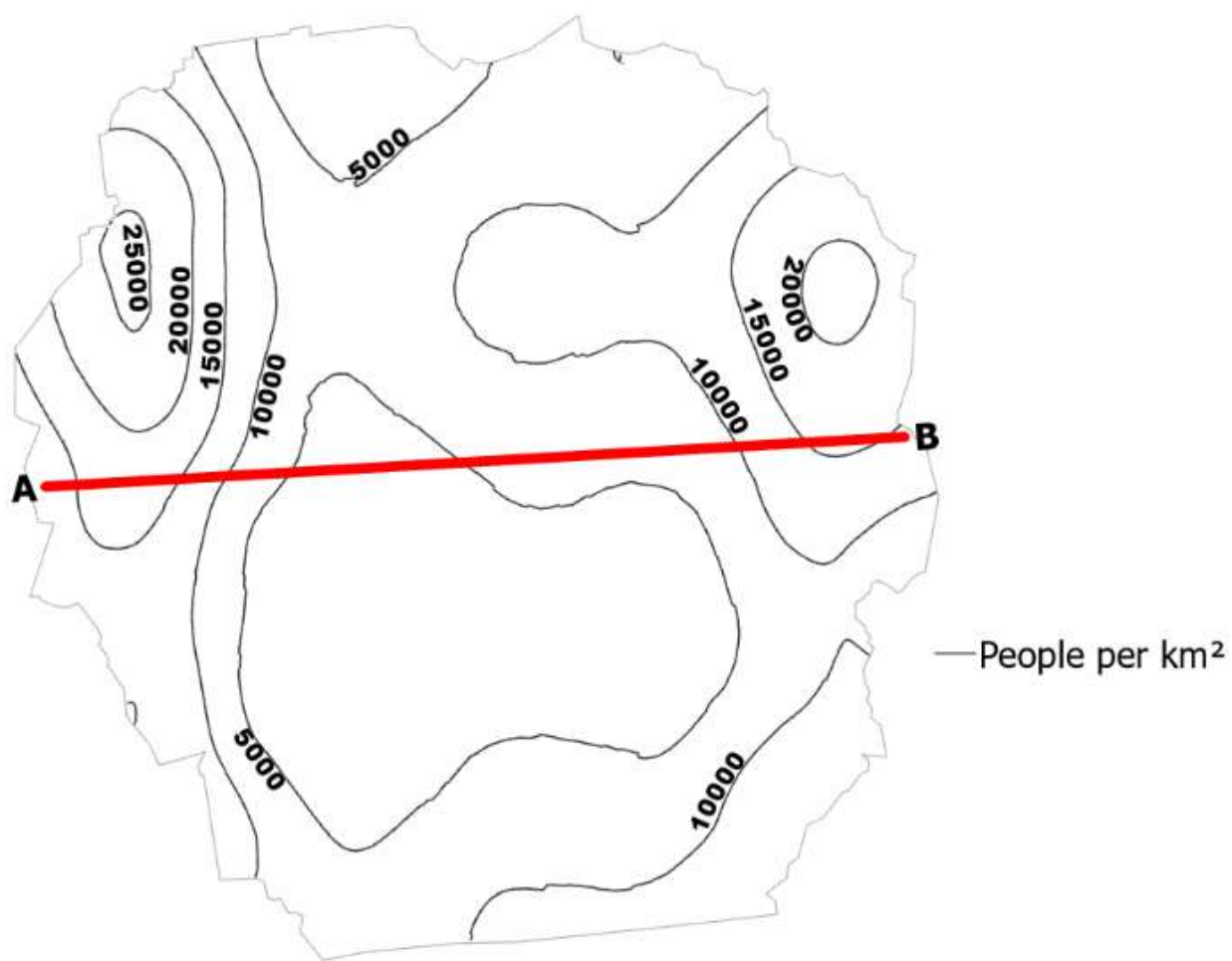




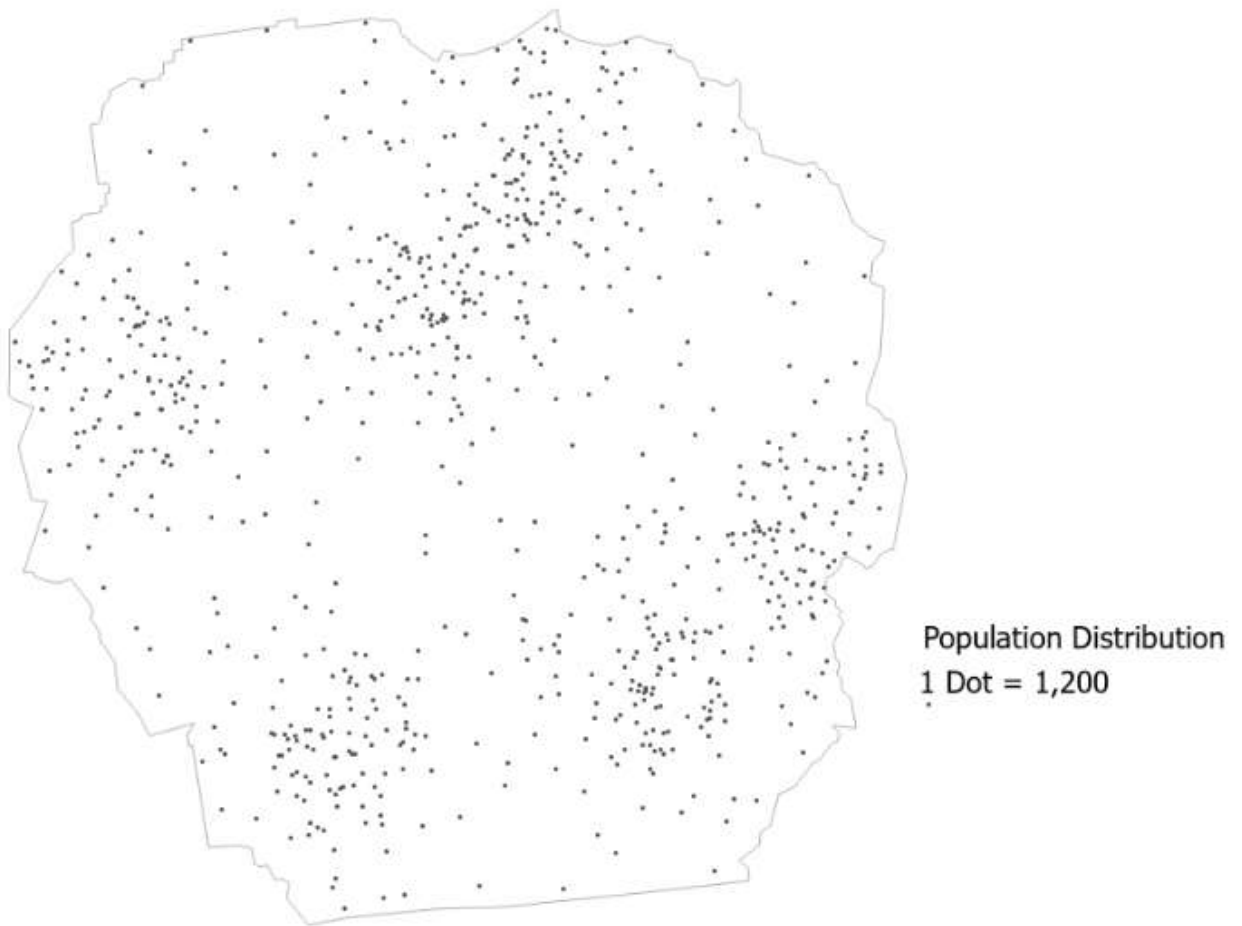
SYN_GD_A

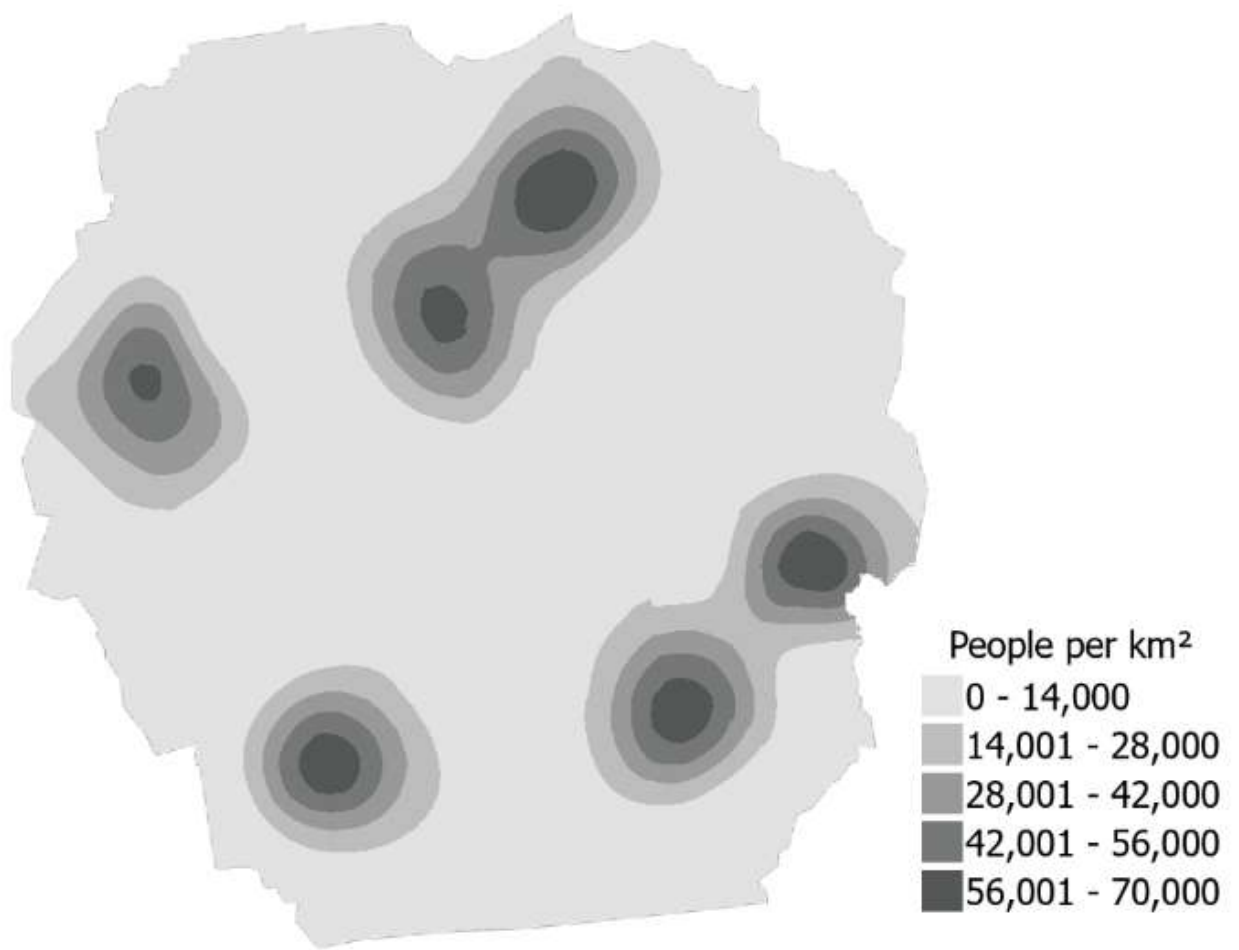


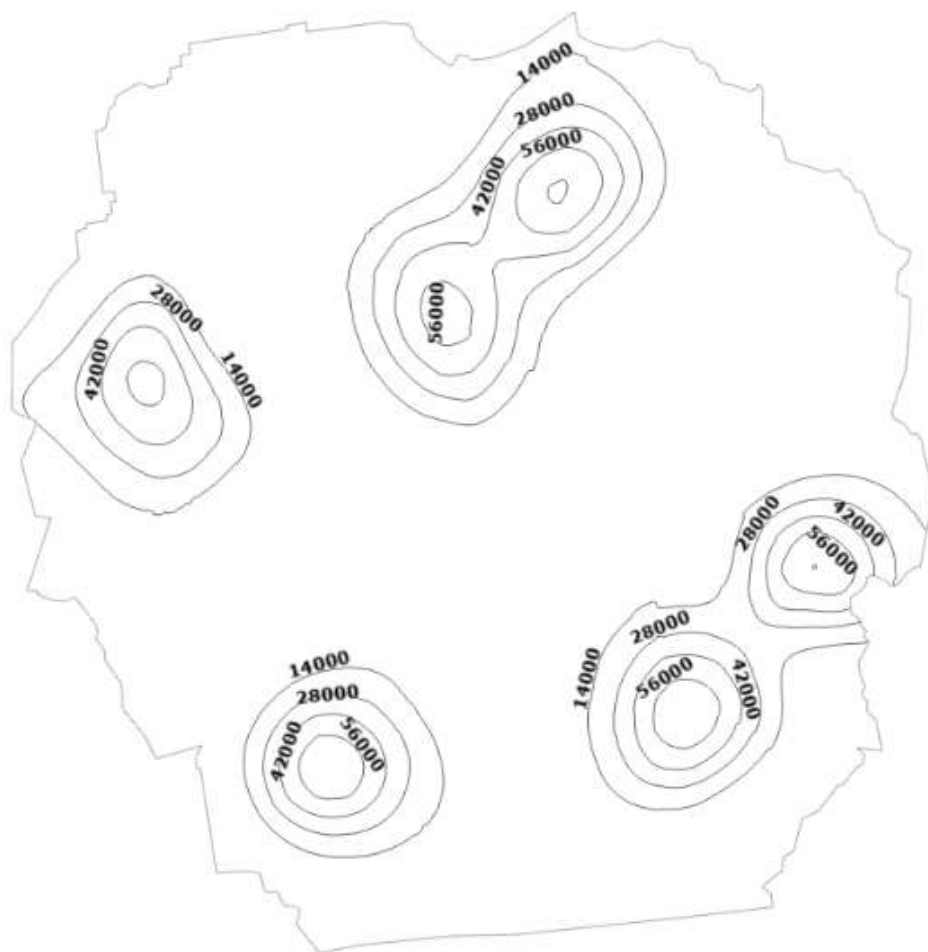




SYN_HS_A

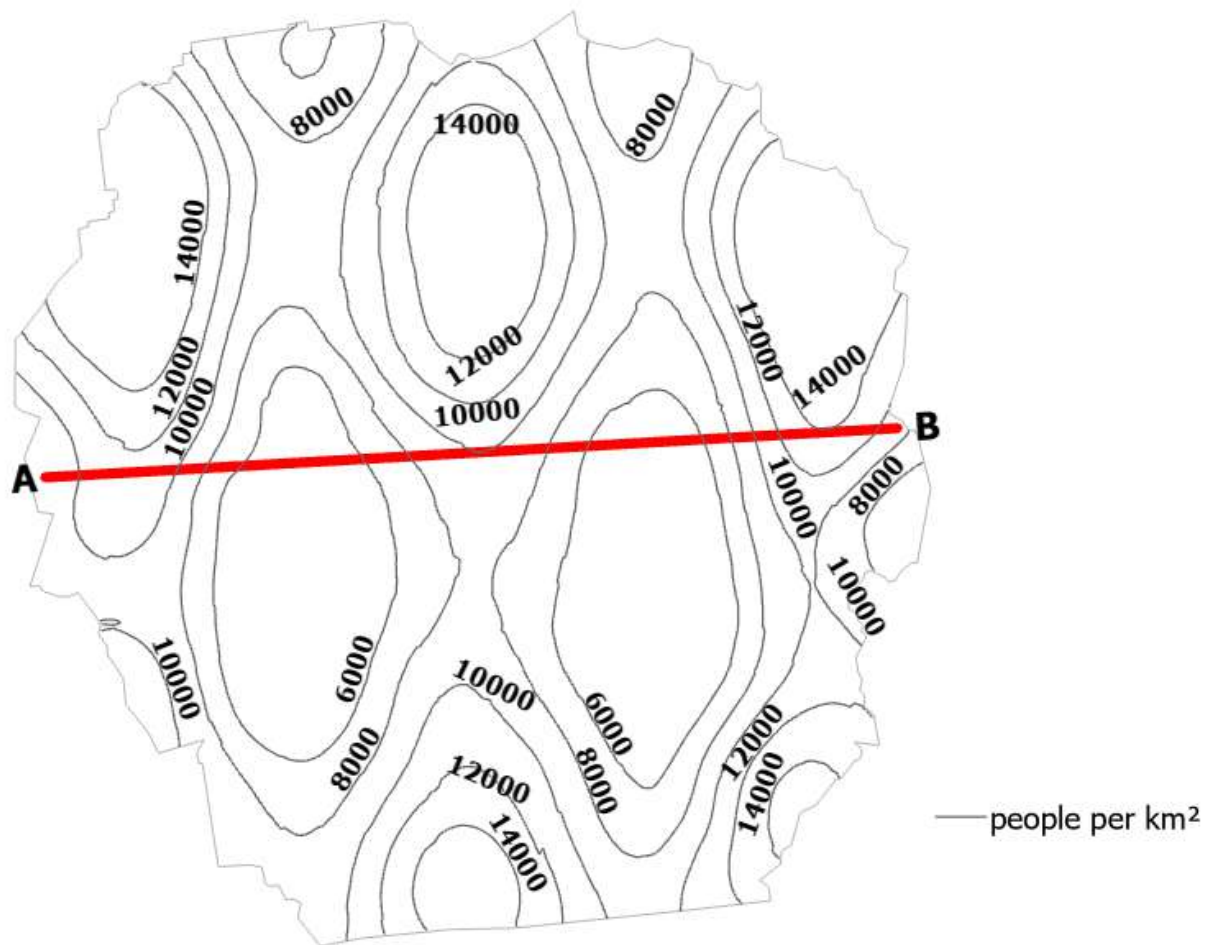


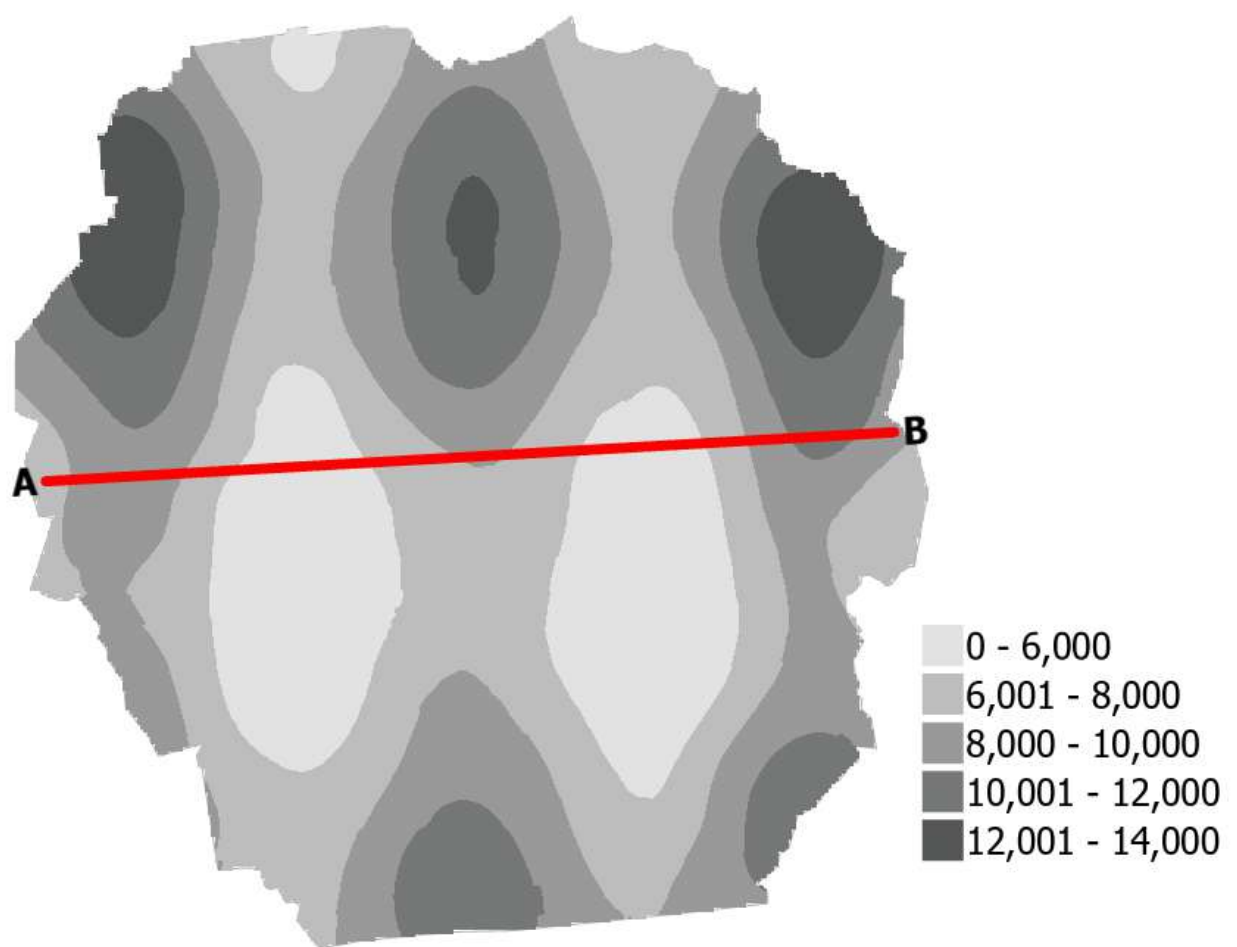


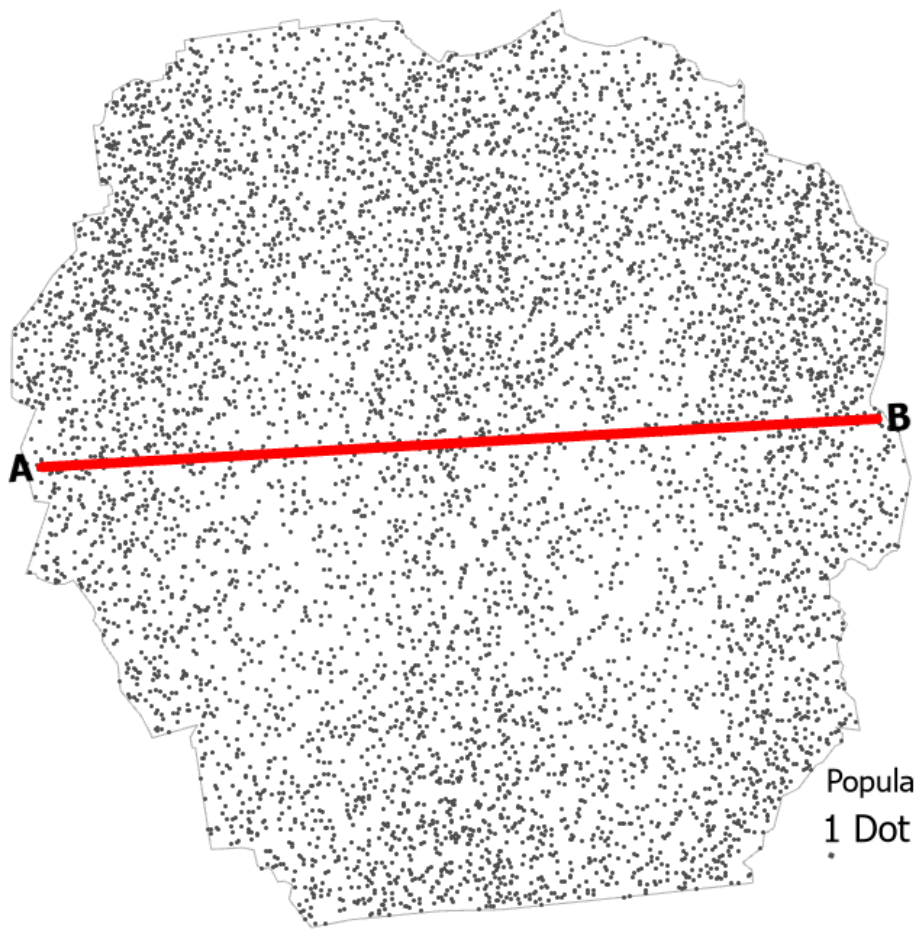


— people per km²

SYN_GD_B







Population Distribution
1 Dot = 100

SYN_HS_B

